UNIT 4 SAMPLING

Structure

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4.1 INTRODUCTION

Sampling has been an age old practice in everyday life. Whenever we want to buy a huge quantity of a commodity, we decide about the total lot by simply examining a small fraction of it. It has been established that the sample survey if planned properly, can give very precise information. Since in surveys a part of the population is only surveyed and inference is drawn about the whole population, the results likely to be different from the population values. But the advantage with the sample survey is that this type of error can be measured and controlled and it can be eliminated to great extent by employing properly trained persons in surveys. The other advantage of sample surveys are that it is less time consuming and involves less cost. Usually, the population is too large for the researcher to attempt to survey all of its members. A small, but carefully chosen sample can be used to represent the population. The sample reflects the characteristics of the population from which it is drawn.

After studying this unit, you should be able to

- discuss the meaning and importance of sampling
- describe the steps and criteria involved in selecting a sampling procedure
- distinguish different types of sampling
- explain the process of determination of sampling size

4.2 SAMPLING: MEANING AND CONCEPT

4.2.1 Meaning of Sampling

According to Levin and Rubin, statisticians use the word, population, to refer not only to people, but, to all items that have been chosen for study. They use the word, sample, to describe a portion chosen from the population.

According to Croach and Housden, a sample is a limited number taken from a large group for testing and analysis, on the assumption that the sample can be taken as representative for the whole group.

According to Boyce, sampling makes an estimate about some of the characteristics of a population. To sample is to make a judgment or a decision about something after experiencing just part of it.

4.2.2 Concepts in Sampling

For clarity and brevity, some concepts and preliminaries of sampling theory, which are used in the study material, are discussed below.

- *Sampling Units and Population:* a unit may be taken as a well defined and identifiable element or a group of elements on which observations can be made. The aggregate of these units is termed as population and the population is said to be finite, if the units are countable. The population is sub-divided into suitable small units known as sampling units for the purpose of sampling. Sampling units may consist of one or more elementary units and each elementary unit belongs to one and one sampling unit.
- *Sampling Frame:* a sampling frame is a list of sampling units with identification particulars indicating the location of the sampling units. A sampling frame represents the population under investigation, and it is the base of drawing a sample. As far as possible, it should be up-to–date, i.e., free from omissions and duplications.
- *Sample:* a fraction of the population is said to constitute a sample. The number of units included in the sample is known as the size of the sample.
- *Sampling Fraction:* the ratio of the sample size, n, to the population size, N, is known as sampling fraction and it is denoted by (n / N).
- *Sampling Procedure/Method:* this is the method of selecting a sample from a population.
- *Census:* this denotes all the elements or units of a population which are used to explain the features of population. It usually refers to complete enumeration of all persons in the population.
- *Population Parameter and Sample Estimator:* any function of the values of units in the population, such as population mean or population variance, is termed a population parameter. There can only be one set of values for a population and the population values are treated as constant. However, the function of the values of the units in the sample, such as sample mean and sample variance, is known as a statistic. The value of the mean and variance differ from sample to sample and, therefore, it is a random variable.

4.2.3 Sampling Advantages

Some of the key advantages of sampling are:

i) it costs less

ii) takes less time

iii) data are acquired quickly

iv) fewer mistakes are likely

v) a more detailed study can be done.

Now that you have read about the meaning and concept of sampling, answer the following questions in *Check Your Progress 1.*

Check Your Progress 1

Note: a) Write your answer in about 50 words.

b) Check your answer with possible answers given at the end of the unit

1) What do you mean by sampling? What are the advantages of sampling?

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2) What is the difference between a parameter and an estimator?

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4.3 TYPES OF SAMPLING

There are broadly two types of sampling:

i) Probability sampling

ii) Non-probability sampling

4.3.1 Probability Sampling

A probability sample is one in which each element of the population has a known, non zero chance of being included in the sample. Probability methods include simple random sampling, systematic sampling, and stratified sampling.

1) Simple Random Sample

The random sample entails that each and every individual in a population has an equal chance of being included in the sample and that the selection of one individual is in no way dependent upon the selection of another person. The two popularly used methods in random sampling are

i) draw of lottery

ii) using a random number table.

i) In lottery draw, for example, if we have to select a sample of 25 students from a total of 600 students in a college, then we make separate slips of paper for 600 students and put them in a box and thoroughly mix them. After that, a person is asked to pick up one slip. Here, the probability of each of the student being selected in the sample is 1/600. This procedure is continued till the sample size is acquired.

ii) Another method of simple random sampling is to use a random number table for drawing 25 students from a total of 600 students. The procedure for using a random number table follows.

1) Number each element in the sample frame from 001 to 600.

2) Decide a random starting point in the table. Any point will do. Say second row in the second column (Appendix 1).

3) Look at the first digits at that point, because there are three digit in 600.

4) Then, if the number is less than 600, include it in the sample; if not then look for a number where the first three digits are less than 600.

5) From that point you can move in any direction. Select only three digit numbers that are less than 600, until you have 25 such numbers.

Note: You can move in any direction in the random number table because every digit has been placed in the table at random.

For example, here if we start from the second row in the third column, then, the random numbers are: 31684; 09865; 14491; 34691, continuing till 25 samples are selected.

2) Systematic Random Sample

Designing a Systematic Random Sample is sometimes quite difficult and time consuming and therefore, Systematic Random Sample, like Simple Random Sample, also uses a list of all members of the population in its sampling frame. However, instead of using random numbers to select the sample elements, the researcher applies a skip interval to the list to produce a sample of the required size.

$$\text{Skip interval} = \frac{\text{number of elements in the population}}{\text{the required sample size}}$$

$$K = \frac{N}{n}$$

K = skip interval
N = Universe size
n = Sample size

For example if we have to select a sample of 100 persons from a universe of 1000 population, then the skip is 10. In this case one number between 1 and 10 has to be selected. Suppose 5 is selected, then the first sample would be 5th and the next one 15th, 25th, 35th, 45th, and so on. One of the advantages of this method is that it is more convenient than other methods and simple to design. Again, it is used with very large populations.

3) Stratified Random Sample

In Stratified Random Sampling, the target population of N units is first divided into k subpopulations of N_1, N_2,....., N_k units. These populations are non-overlapping and together they comprise the whole population. So that $N_1 + N_2 + ... + N_k = N$ respectively If the total sample size 'n' is to be drawn from the target population than

$n_1 + n_2 + ... + n_k = n$

If a simple random sample is drawn in each stratum, the whole procedure is described as stratified random sampling.

Stratified random sampling requires more than making a list of elements (and estimating the number of elements on the list). It also involves ordering that list by sub groups (or strata) and then, to do sampling randomly or systematically within those sub groups. This method of sampling is used for the following reasons.

- It can reduce the errors in the statistical estimates calculated from the sample.
- It allows you to create a sample that is exactly representative of the various sub groups in the population that you find to be of special interest.

For example, the selected village may have households of SC, ST, OBCs, Others, Minority. The village population first may be divided in to smaller sub groups of different sections of population (stratum) and, thus, the village sample may consist of households from each stratum so that sample may contain all the important characteristics of the village population. In the case of SRS, the sample of all strata/ sub groups sometimes may not be included or covered adequately.

- This method helps in conducting and managing a large scale survey to be conducted in a country like India. The agency conducting the survey may have field offices in different locations; each one can supervise the survey for a part of the population.
- The basic idea is that it sub-divides the heterogeneous population into homogeneous sub-populations. If each stratum is homogenous in itself, a precise estimate of any stratum mean can be obtained from a small sample, thus, saving a lot of time and cost.

There are two types of stratified samples.

A **proportionate stratified sample** selects the number of elements from each stratum so that the stratum sample size ($n_1, n_2, \ldots\ldots\ldots\ldots, n_k$) is proportional to their respective stratum population size ($N_1, N_2, \ldots\ldots\ldots\ldots, N_k$).

Consider the following examples:

- A selected village may have households of SC(10%), ST (5%), OBCs (45%), Others (30%), Minority (10%). A village sample of 100 may constitute the households of various casts in the above proportion/percentage so that the sample may contain all important characteristics of village population.

- Hospital patients are stratified according to age, dividing the population into those who are aged 50years or above, and, those who are under 50. If there are twice as many people aged 50 or above admitted to the hospital as those under 50, a proportionate stratified sample will include twice as many people aged 50 or above.

A **disproportionate stratified sample** selects the number of elements from each stratum so that the stratum sample size is not proportional to the stratum population size. The most common reason for selecting this type of sample is when you want to study a relatively rare but important subpopulation, such as younger patients suffering from heart disease. Proportionate stratification may result in too few elements being selected so that little, if any, statistical analysis

can be done. Consequently, even if these patients represent only 1% of the population, you might decide to make them 10% of the final sample. However, once we combine values of all stratums, the size of the higher selected proportion needs to be readjusted which is called weighted estimate.

4) Probability Proportion to Size (PPS) Sample

It has been observed that the elementary units of the population vary in size. Such ancillary information about the size of the unit can be utilized in selecting the sample so as to get better and efficient estimates of the population parameter. For example villages with larger geographical area are likely to have larger area under food crops; therefore, in estimating the production, it would be desirable to adopt a sampling scheme in which villages are selected with probability proportional to geographical area. When units vary in their size and the variable under study is directly related with the size of the unit, the probabilities may be assigned proportional to the size of the unit.

Probability Proportion to Size (PPS) Sampling assures higher probability of selection to sampling unit which are larger in size. This technique was initially used in estimation of crop production, fruits production etc because productivity is directly related with the size of field. In social science surveys also characteristics of village population is influenced by the size of population. The **procedure of selecting the sample** is described below.

Suppose you have to select 5 villages from the list of 10 using PPS sampling. First arrange all villages in ascending or descending order of population size as may be seen in column 2 of the table 1. Then, in the third column, find the cumulative sum of population size and in the fourth column, assign them range of serial numbers as shown below in the table.

Table 4.1: Village population Size

Serial Number	Village Population Size	Cumulative Sum of Population Size	Cumulative Population Size Interval
1	2	3	4
1	200	200	0001 - 0200
2	250	450	0201 - 0450
3	300	750	0451 - 0750
4	350	1100	0751 - 1100
5	400	1500	1101- 1500
6	450	1950	1501 - 1950
7	500	2450	1951 - 2450
8	550	3000	2451 - 3000
9	600	3600	3001 - 3600
10	650	4250	3601 - 4250
Total	4250		

Please notice that the total population of all villages in the target population is a four digit number (4250). Therefore, initially, a random number in four digits, which is less than or equal to the total population of all villages (4250), is selected from the random number table. For example, it is 0331 which will correspond to serial number 2. Next random number is 4320; therefore, it may be discarded. The next number selected is 1296; therefore, it will correspond to serial number 5. The next random numbers may be 1553, 2402 and 3640 which will correspond to serial numbers 6, 8, and 10 respectively. In this way, selected villages will be serial numbers 2, 5, 6, 8, 10.

5) Cluster Sample

Cluster sampling is a sampling technique used when natural groupings are evident in a statistical population. It is often used in marketing research. In this technique, the total population is divided into these known groups (or clusters) and a sample of the groups is selected. Then the required information is collected from the elements within each selected group. This may be done for every element in these groups, or a sub sample of elements may be selected within each of these groups. The technique works best when most of the variation in the population is within the groups, not between them.

Briefly, the procedure for selecting a cluster sample is given below.

- The population is divided into N groups, called clusters.
- The researcher randomly selects n clusters to include in the sample.
- The number of observations within each cluster is known:

  $$M = M1 + M2 + M3 + \ldots\ldots\ldots + MN$$

- Each element of the population can be assigned to one, and only one, cluster.

Cluster sampling should be used only when it is economically justified - when reduced costs can be used to overcome losses in precision. This is most likely to occur in the following situations.

- Constructing a complete list of population elements is difficult, costly, or impossible. For example, it may not be possible to list all elementary units of the populations, for example all households in village, block, etc. However, it would be possible to randomly select a subset of villages, blocks (stage 1 of cluster sampling) and, then, interview the head of family in a house of the selected cluster (stage 2).

- The population is concentrated in natural clusters (city blocks, schools, hospitals, etc.). For example, to conduct personal interviews of operating room nurses, it might make sense to randomly select a sample of hospitals (stage 1 of cluster sampling) and then interview all of the operating room nurses at that hospital. Using cluster sampling, the interviewer could conduct many interviews in a single day at a single hospital. Simple random sampling, in contrast, might require the interviewer to spend all day travelling to conduct a single interview at a single hospital.

As discussed above, in the cluster sampling method, the primary selecting unit is not a household, rather a natural cluster of households, viz., hamlets in villages, or, created clusters, viz., schools, malls, etc., may be decided. The first list of clusters may be selected using the SRS or the PPS sampling techniques. Then,

from each selected cluster, all units, or, some of the units, may be selected as per the required sample size using Stratified Random Sampling or the Systematic Random Sampling techniques.

This sampling technique is quite popular in evaluation surveys in health – it is also called the 30 Cluster Sampling Technique. This is also a rapid method of data collection as the researcher can collect more data in less time due to the decrease in transportation time as compared with other sampling techniques.

4.3.2 Non-Probability Sampling

A **non-probability sample** is one in which a case in a sample is chosen in such a manner that it gives you information for the sample itself and makes it possible to generalize the findings for the population with certain degree of precision. Such a sample is also called a purposive sample. This kind of sampling is primarily used to collect information on market surveys to know the attitude, opinion, behaviour, reactions of individuals. There are many types of non-probability samples, including snowball sampling, convenience, purposive/ judgment, quota sampling, etc.

1) Convenience Sample

The convenience sample is so called because it is relatively easy to obtain and contact. In this method the investigators are usually asked to select the people for the interview in accordance to the instructions from the researcher. The benefit of a convenience sample is that the interviewer can usually get interviews done quickly and cheaply. Convenience sampling is appropriate for exploratory research.

2) Judgments Sample

A judgment sample is similar to that of convenience sample. In a judgment sample, the researcher selects samples that are believed to represent the population. The selection of samples is based on the knowledge of the population and the characteristics which the sample is to represent. It is less costly and very useful for forecasting.

3) Quota Sample

Quota sampling is like stratified sampling. In quota sampling, the population is categorized into several strata which consist of an expected size, and the samples are considered to be important for the population they represent. The advantages of quota sample are that it involves a short time duration, is less costly, and gives moderate representation to a heterogeneous population.

4) Snowball Sample

This is one of the important types of non-probability sampling. In snowball sampling, the investigator encourages the respondents to give the names of other acquaintances and it continues growing in size and chains until the research purpose is achieved. It is also, therefore, known as networking, chain, or referred sampling method. It is very useful in the study of networking and is less costly.

A comprehensive overview of the various types of sampling can be seen in figure 4.1

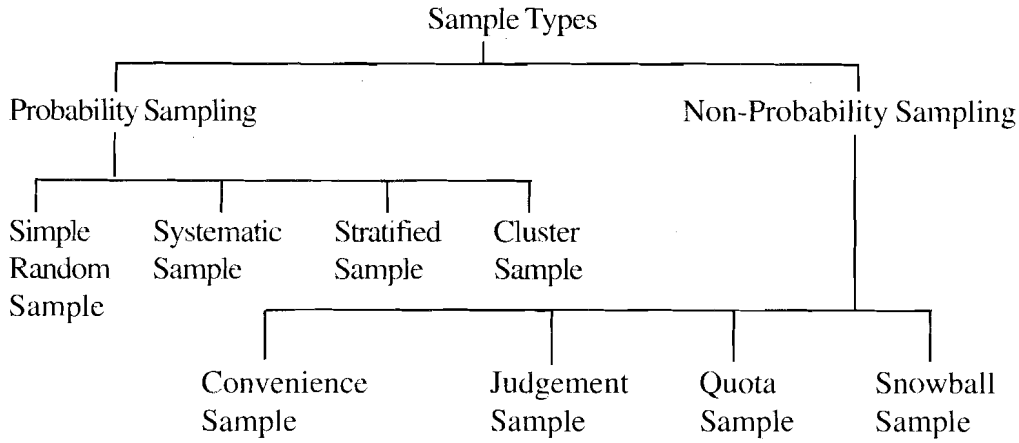


Fig-4.1: Types of Sample

4.4 SAMPLE DESIGN PROCESS

The sample design process follows five steps as given in Box-1

Box 1 Sample Design Process

Step-1	Define the Population
Step- 2	Decide if it would be better to take a sample or a census
Step-3	If possible, construct or obtain an appropriate sampling frame
Step-4	Decide whether to use a probability sample or a non-probability sample
Step- 5	Select the sampling method

Source: John Boyce (www.mhhe.com/av/boyceze)

Step 1: Define the Population

We use the word, population, frequently in our day-to-day conversations, for example, 'The population of India', or, 'The population of Punjab', or, 'The population of Kerela'. However, the meaning of the word, population, in research is different from what we use in day-to-day conversation. A research population may be defined as 'a clearly defined group of entities that have some characteristics in common'. This means the kind of people on whom we wish to base our research project. Sometimes, in research, we use the word, universe, instead of population.

In a research project, our intention is to learn or infer something about the population. Whether we would use a sample or a population has to be clearly defined. For example, if we want to conduct a study on road safety, then the task of defining the population for a survey would be whether we should

- interview only the people who drive two wheelers
- interview only the people who drive four wheelers
- interview the pedestrians
- interview only who are hand rickshaw pullers or ride bi-cycle.

Therefore, judging a population is the starting of the sampling process.

Step- 2: Decide whether to take a sample or a census

After judging the population, the next step in the sampling process is to decide whether to take a sample or a population in your research project. In a census, usually every member of the population is interviewed. While in a sample method only selected members of the population are included. From the census we obtain data that are called population parameter, and from the sample we obtain statistics in a parameter. A parameter is a measurement of a characteristic of a population, while a sample statistic is used as an estimate of a population parameter.

Usually, in individual research, we use census when the population is small, and sample when the population is large.

Step-4: Decide whether to use a probability sample or a non-probability sample

The fourth step in the sampling process is whether to use probability sampling or non-probability sampling. In a probability sample, the sample elements are chosen by random selection, while in non-probability sampling, each sample element is chosen according to whether the researcher decides that it should be included or not.

Step- 5: Select the sampling method

Last, but not least in the sampling process, is the selection of the sampling method. In the probability sampling method, the following four principal kinds of probability sampling are used: the simple random sample, the systematic sample, the stratified sample, and the cluster sample. The main non-probability sampling methods are the convenience sample, the judgment sample, the quota sample, and the snowball sample.

Sampling Frame

A sampling frame is a listing of all the elements from which you will draw the sample. In the ideal situation, the sampling frame will include all elementary units in the target population. A list of employees in an organization can create a sampling frame that exactly matches the population of interest. You should try to ensure that the sampling frame has the following characteristics.

- It is actually created from the target population.
- It is as complete a list as possible of the elements in the population.

In this section, you have read about the various types of sampling and the sample design process. Now, answer the following questions in *Check Your Progress 2.*

Check Your Progress 2

Note: a) Write your answer in about 50 words.

b) Check your answer with possible answers given at the end of the unit

1) What is a stratified sample?

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2) What is a cluster sample and when is it best used?

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3) What do you understand by a quota sample?

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4.5 ERRORS IN SAMPLING

Many mistakes and errors in social science research happen because of misleading and biased sampling. A sample which does not represent the population is called a biased sample. According to Yule and Kendal, "Bias may be due to imperfect instruments, the personal qualities of the observer, defective techniques and other cases. Like experimental error, it is difficult to eliminate entirely, but usually may be reduced to relatively small dimensions by taking proper care." There are two types of errors such as sampling errors and non-sampling errors. These are discussed below:

i) Sampling Error

By definition, when you have collected a sample from a population, you have less than complete information about the population. This, in turn, means that there is a chance that the sample statistics you calculate, (for example, the mean of a variable, a frequency distribution, etc.) may not be an unbiased estimate of the population parameter.

The error in the sample estimate is not an intrinsic impediment to analysis. For probability samples, sampling theory allows you to calculate the expected amount of error given a particular sample size, sampling method, and the specific statistic of interest. In general terms, the sampling error for a statistic can be defined as:

$$\text{Standard error} = \frac{sd}{\sqrt{n}}$$

Where n refers to the number of respondents (sample size).

As the sample size increases, the standard error of a statistic decreases; as the variance, or dispersion, of a statistic increases, so does its sampling error.

Sampling error decreases rapidly as the sample size increases from a few hundred to about 1000 respondents. However, there is rarely any reason to select larger samples while comparing the increased cost of survey with reduction in sampling error (see 'Calculating the Sample Size', in next section).

The formula for the standard error of a proportion is simple and easy to apply:

$$\text{Standard error} = \sqrt{\frac{p \times (1-p)}{n}}$$

Here, p represents the proportion of successes (favourable response, those who received the benefits), {q = (1-p)} represents the proportion of failures (those who did not receive the benefits), and n is the total number of respondents. The standard error of a statistic is greatest when p and (1-p) are equal, which occurs when each is 0.50, or 50%, of the sample.

ii) Non-Sampling Error

Before discussing how to determine sample size, we will briefly review other sources of error in surveys. When you read a a news article that reports the results of a national poll, the error in the estimates is always listed, derived, generally speaking, from Equation 6.2. However, experienced survey researchers know that errors due to other sources are typically greater than the error due to sampling alone. Following are some other types of errors.

- Measurement errors, caused by poorly written questions, poorly designed questionnaires, respondent errors in completing questionnaires, and so on.
- Non-response errors, caused because the respondents are not a representative subset of the population.
- Data coding errors, caused, by errors in coding and entering the data.

Of these error sources, the first two are typically more severe. In mail surveys, non-response error is often the most serious problem.

There are two critical characteristics of these non sampling errors. First, as mentioned above, their sum is often greater than the sampling error. Second, and more insidious, these errors are often impossible to estimate for any one survey, especially measurement and non-response errors. Consequently, using Equation 6.1 and Equation 6.2 to estimate the error in a statistics often provides a false sense of security.

Experienced survey researchers take this fact into account by being more cautions in discussing survey results than the sampling error alone would indicate, and you should do the same. Ideally, the other sources of error would balance

themselves out so that errors in one direction negate errors in the other directions, but you cannot assume that this is the case.

4.6 DETERMINATION OF SAMPLE SIZE

The sample size can be determined by:

i) Using a formula

ii) Using a table

4.6.1 Determining Sample Size Using a Formula

$$n = \frac{Z^2pq}{d^2}$$ (when population is greater than 10,000)

$$nf = \frac{n}{1+(n/N)}$$ (when population is less than 10,000)

n, nf = desired sample size

Z = the standard normal deviate

p = the portion in the target population estimated to have a particular characteristic. If there is no reasonable estimate, then use 50 percent (.50).

q = 1-p

d= degree of accuracy desired, usually set at .05 or occasionally at .02.

n = the estimate of the population size

Z at 99% confidence level i.e. at 1% level of significance = 2.58

Z at 95% confidence level i.e. at 5% level of significance = 1.96

Z at 90% confidence level i.e. at 10% level of significance = 1.65

Exaple: (when population is more than 10,000)

If the proportion of target population with a certain characteristic is .50, the Z statistic 1.96 and we desire accuracy at the 0.05 level, then the sample size is

$$n = \frac{(1.96)^2\ (.50*.50)}{(0.05)^2}$$

$$= \frac{3.84*0.25}{(.0025)}$$

$$= \frac{0.96}{0.0025}$$

$$= 384$$

If we use the more convenient 2.0 for the Z statistic, then the sample size will be smaller.

$$nf = \frac{400}{1 + \frac{(400)}{1000}}$$

$$= \frac{400}{1.4}$$

$$= 286$$

4.6.2 Determining Sample Size by Using a Table

Another way to determine sample size is to rely on published tables which provide the sample size for a given set of criteria. Table 1 presents sample size values that will be appropriate for many common sampling problems. The table includes sample sizes for both continuous and categorical data assuming alpha levels of .10, .05, or .01.

Table 4.2: Table for Determining Minimum Returned Sample Size for a Given Population Size for Continuous and Categorical Data

Population size	Sample size					
	Continuous data (margin of error= .03)			Categorical data (margin of error= .05)		
	alpha= .10 t=1.65	alpha= .05 t=1.96	alpha= .01 t=2.58	p=.50 t=1.65	p=.50 t=1.96	p=.50 t=2.58
100	46	55	68	74	80	87
200	59	75	102	116	132	154
300	65	85	123	143	169	207
400	69	92	137	162	196	250
500	72	96	147	176	218	286
600	73	100	155	187	235	316
700	75	102	161	196	249	341
800	76	104	166	203	260	363
900	76	105	170	209	270	382
1,000	77	106	173	213	278	399
1,500	79	110	183	230	306	461
2,000	83	112	189	239	323	499
4,000	83	119	198	254	351	570
6,000	83	119	209	259	362	598
8,000	83	119	209	262	367	613
10,000	83	119	209	264	370	623

In this session you studied about errors in sampling and determination of sample size. Now, answer the questions given in *Check Your Progress 3.*

Check Your Progress 3

Note: a) Write your answer in about 50 words.

b) Check your answer with possible answers given at the end of the unit

1) What is sampling error?

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2) How is the sample size determined using the formula?

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4.7 LET US SUM UP

In this unit, we discussed the meaning and various concepts in sampling particularly of sample and population. There is also a detailed discussion on the sample types and sample design process. There are two types of sampling such as probability and non-probability sampling. The types of probability sampling are the Simple Random Sample, the Systematic Sample, the Stratified Sample, and the Cluster Sample, while different types of non-probability sample are the Convenience Sample, the Quota Sample, the Judgment Sample, and the Snowball Sample. The unit also discusses various steps of the sampling design process. This is followed by two of the very important concepts of sampling: the determination of sample size and errors in sampling.

4.8 KEYWORDS

Sample : a sample is simply a subset of a larger aggregation, i.e., typically a population and it contains all the characteristics of a population,

Sampling : **the** process of selection of subjects/study elements to create a sample for collecting information about a population.

Standard Error : this is the expected amount of error while estimating the specific statistic of interest, using a particular sample size and sampling method with respect to actual population value.

Sampling Error : while collecting information from a sample, there is a chance that the sampling statistics may not be equal to the same values in the population. The error is that the sample does not contain complete information about the population.

Confidence Interval : this gives the probability of the sample estimate falling within the interval.

Sample Size : **the** number of elementary units in a sample is called a sample size.

2711	8182	75997	79866	58095	83319	80295	79741	74599	84379
94873	90935	31684	63952	9865	14491	99518	93394	34691	14985
54921	78680	6635	98689	17306	25170	65928	37709	30533	89736
77640	97636	37397	93379	56454	59818	45827	74164	71666	46977
61545	835	93251	87203	36759	49197	85967	1704	19634	21898
17147	19519	22497	16857	42426	84822	92598	49186	88247	39967
13748	4742	92460	85801	53444	65626	58710	55406	17173	69776
87455	14813	50373	28037	91182	32786	65261	11173	34376	36408
8999	57409	91185	10200	61411	23392	47797	56377	71635	8601
78804	81333	53809	32471	46034	36306	22498	19239	85428	55721
82173	26921	28472	98958	7960	66124	89731	95069	18625	92405
98594	25168	89178	68190	5043	17407	48201	83917	11413	72920
73881	67176	93504	42636	38233	16154	96451	57925	29667	30859
46071	22912	90326	42453	88108	72064	58601	32357	90610	32921
44492	19686	12495	93135	95185	77799	52441	88272	22024	80631
31984	72170	37722	55794	14636	5148	54505	50113	21119	25228
51574	90692	43339	65689	76539	27909	5467	21727	51141	72949
35350	76132	90925	92124	92634	35681	43690	89136	35599	84138
46943	36502	1172	46045	46991	33804	80006	35542	61056	75666
22665	87226	33304	57975	3985	21566	65796	82915	81466	89205
39437	97957	11838	10433	21564	51570	73558	27495	34533	57808
77082	47784	40098	97962	89845	28392	78187	6112	8169	11861
24544	25649	43370	28007	6779	72402	62632	53956	24709	6978
27503	15558	37738	24849	70722	71859	83736	6016	94397	12529
24590	24545	6435	52758	45685	90151	45616	49644	92686	84870
48155	86226	40359	28723	15364	69125	12609	57171	86857	31702
20226	53752	90648	24367	83314	14	19207	69413	97016	86290
70176	73444	38790	53626	93780	18626	68766	24371	74639	30782
10169	41465	51935	5711	9799	79077	88159	33437	68519	3040
81084	3701	28598	70013	63794	53169	97054	60303	23259	96196
69202	20777	21727	81511	51887	16175	53746	46516	70339	62727
80561	95787	89426	93325	86412	57479	54194	52153	19197	81877

8199	26703	95128	48599	9333	12584	24374	31232	61782	44032
98883	28220	39358	53720	80161	83371	15181	11131	12219	55920
84568	69286	76054	21615	80883	36797	62845	39139	90900	18172
4269	35173	95745	53893	86022	77722	52498	84193	22448	22571
10538	13124	36099	13193	37706	44562	57179	44693	67877	1549
77843	24955	25900	69843	95029	93859	93634	20205	66294	41218
12034	94636	49455	76362	83532	31062	69903	91186	65768	55949
10524	72823	47641	93315	80875	28090	97728	52560	34937	79548
68935	76632	46984	61772	92786	22651	7086	89754	44143	97687
83450	65665	29190	43709	11172	34481	95977	47535	25658	73898
90696	20451	24211	97310	60446	73530	62865	96574	13829	72226
49006	32047	93086	112	20470	17136	28255	86328	7293	38809
74591	87025	52368	59416	34417	70557	86745	55809	53628	12000
6315	17012	77103	968	7235	10728	42189	33292	51487	64443
62386	9184	62092	46617	99419	64230	96034	85481	7857	42510
86848	82122	4028	36959	87827	12813	8627	80699	13345	51695
65643	69480	46598	4501	40403	91408	32343	48130	49303	90689
11084	46534	78957	77353	39578	77868	22970	84349	9184	10603

4.9 REFERENCES AND SELECTED READINGS

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4.10 CHECK YOUR PROGRESS – POSSIBLE ANSWERS

Check Your Progress 1

1) What do you mean by sampling? What are the advantages of sampling?

Answer: Generally, a sample implies a small representative of a large whole. This sampling method is frequently used in social science research to save time. Some of the key advantages of sampling are: (i) it costs less; (ii) takes less time; (iii) data are sometimes wanted quickly; (iv) fewer mistakes are likely; (v) a more detailed study can be done.

2) What is the difference between a parameter and an estimator?

Answer: Any function of the values of units in the population, such as the population mean or population variance, is termed, a population parameter. There can only be one set of values for a population, there population values are treated as constant. However, the function of the values of the units in the sample, such as the sample mean and sample variance is known as a statistic. The value of the mean and variance differs from sample to sample and, therefore, it is a random variable.

Check Your Progress 2

1) What is stratified sampling?

Answer: In *stratified sampling*, the target population of N units is first divided into k subpopulations of N_1, N_2,....., N_k units. These populations are non-overlapping and together they comprise the whole population, so that

$N_1 + N_2 + ... + N_k = N$

The sub-populations are called strata. The number in each stratum should be known. A sample is drawn from each stratum independently. The sample sizes within 'k' strata are denoted by $n_1, n_2, ... , n_k$ respectively If the total sample size n is to be drawn from the target population then

$n_1 + n_2 + ... + n_k = n$

If a simple random sample is drawn in each stratum, the whole procedure is described as *stratified random sampling.*

2) What is cluster sampling and when is it best used?

Answer: Cluster sampling is a sampling technique used when natural groupings are evident in a statistical population. It is often used in marketing research. In this technique, the total population is divided into these known groups (or clusters) and a sample of the groups is selected. Then, the required information is collected from the elements within each selected group. This may be done for every element in these groups or a sub sample of elements may be selected within each of these groups. The technique works best when most of the variation in the population is within the groups, not between them.

3) What do you understand by quota sample?

Answer: Quota sampling is like stratified sampling. In quota sampling, the population is categorized into several strata which consist of an expected size and they are considered to be important for the population they are supposed to represent. The advantages of the quota sample are: shorter time duration, less costly, and gives moderate representation to a heterogeneous population.

Check Your Progress 3

1) What is sampling error?

Answer: By definition, when you have collected a sample from a population, you have less than complete information about the population. This, in turn, means that there is a chance that the sample statistics you calculate, (for example, the mean of a variable, a frequency distribution, etc.) may not be unbiased estimate of the population parameter. This error is called sampling error.

2) How is the sample size determined using the formula?

Answer: The calculation of the sample size is concerned with the number of respondents required. To determine the number to select for the sample drawn from the sampling frame, you must estimate the non-response rate. The actual sample size to be drawn is:

$$Sample\ size = \frac{Number\ of\ respodents}{Response\ rate}$$

So, if any survey organization decides that they need 700 respondents, and the expected response rate from the population is 50%, then 700/0.50, or 1400, customers must be drawn from the sampling frame.

UNIT 1 TYPES OF DATA

Structure

1.0 Introduction
1.1 Objectives
1.2 Types of Data: Quantitative and Qualitative
1.3 Quantitative Data
- 1.3.1 Types of Quantitative Data
- 1.3.2 Tabulation and Organisation of Quantitative Data
  - A. Frequency Distribution
  - B. Cumulative Frequency Distribution
- 1.3.3 Graphical Presentation of Quantitative Data
- 1.3.4 Analysis of Quantitative Data
  - A. Measures of Central Tendency
  - B. Measures of Variability
  - C. Measures of Relative Position
  - D. Measures of Relationship

1.4 Qualitative Data
- 1.4.1 Organisation of Qualitative Data
- 1.4.2 Analysis of Qualitative Data
  - A. Content Analysis
  - B. Inductive Analysis
  - C. Logical Analysis

1.5 Let Us Sum Up
1.6 Glossary
1.7 Check Your Progress: The Key

1.0 INTRODUCTION

In Block 3, we dealt with the nature of various tools used in the collection of data. These data are mostly expressed in quantified terms. However, quantitative data may not be available in certain cases. In such a situation, the researcher has to consider the phenomenon as a whole without breaking it down into measurable or quantifiable components. Indeed, he/she should be familiar not only with the two types of data – quantitative and qualitative, but also with the process of classifying data, graphical representation and the various methods of data analysis.

The aim of this Unit is to make you understand the nature of quantitative and qualitative data, the procedures for classifying and tabulating quantitative data, presenting them graphically, and the various methods used in data analysis.

1.1 OBJECTIVES

After the completion of this Unit, you should be able to:

- Define quantitative and qualitative data,
- Prepare various types of graphs and tables for presenting data,
- Compute measures of central tendency, variance, standard deviation, measures of relative position and measures of relationships, and
- Describe various methods used for analysing qualitative data.

1.2 TYPES OF DATA: QUANTITATIVE AND QUALITATIVE

The data collected through the administration of various types of tools on the selected samples are of two types - **qualitative** and **quantitative**. In **quantitative data**, numerical values are assigned to the characteristics or properties of objects or events, according to logically accepted rules. It is a process wherein a number system like figures, ratings or scores is imposed on empirical data. However, when the researcher takes into consideration the phenomenon as a whole and does not attempt to analyse it in measurable or quantifiable terms, the approach becomes **'qualitative'**. Generally, in educational and behavioural sciences, both types of data, (i) quantitative and (ii) qualitative, are recognised. We will look at the characteristics of both in the following sections. Section 1.3 deals with quantitative data, their types, tabulation, frequency distribution and cumulative frequency distributions, the need to represent data graphically, the various types of graphs, and the methods of analysing quantitative data, viz., measures of central tendency, variability, relative positions and relationship. In section 1.4, we shall briefly discuss qualitative data and their analysis with reference to content analysis, logical analysis and inductive analysis. The application of various parametric and non-parametric tests is discussed in more detail in Unit 2 of this Block.

1.3 QUANTITATIVE DATA

Quantitative data describe an empirical event or phenomenon in a numerical system with the help of different scales of measurement: **nominal, ordinal, interval** or **ratio**. **Nominal scales** of measurement are used when a set of objects among two or more categories is to be differentiated on the basis of certain clearly known characteristics. For example, we may assign individuals categories such as sex (male and female), nationality (French and Indian), level (school and higher), education (formal and non-formal), etc. The **ordinal scales** of measurement correspond to quantitative classification of a set of objects done with the help of ranking on a continuum. For example, we may rank students according to their height from the lowest to the highest. The **interval scale** of measurement is based on equal units of measurement.

It includes how much or how little of a given characteristic or attribute is present. For example, the difference in the amount of an attribute possessed by individuals with test

scores of 60 and 61 is assumed to be equivalent to that between individuals with scores of 50 and 51. **Ratio scale** is the highest level of measurement. Since this scale assumes the existence of absolute zero, this type of measurement is almost non-existent in educational and psychological measurement. Thus, quantitative data are expressed in nominal, ordinal, interval and ratio scales of measurement.

1.3.1 Types of Quantitative Data

Quantitative data may be classified into two categories:

(i) Parametric and
(ii) Non-parametric

Parametric data are obtained by applying interval or ratio scales of measurement. Scores of the tests of ability, achievement, attitude, interest, values, personality etc. are examples of interval scales of measurement. In the study of reaction time we use ratio scale. In this type of experiment, the zero point in the absolute sense is known and it makes sense to look at the ratio of the time taken to respond in different treatment situations.

Non-parametric data are either counted or ranked. In counted data, we make use of the nominal scale. Each individual can be a member of only one category and all the members of that category have the same, defined characteristics. For example, we may categorise a group as a sample of 'female students' of a particular 'study centre' of an open university. The categorisation of teachers at different educational levels—school, college and university—is another example of nominal data.

In ranked data, we apply the ordinal scale of measurement. The sets or classes of objects are ordered on a continuum in a series ranging from the lowest to the highest according to the characteristics we wish to measure. The ranking of students in a class for height, weight or academic achievement are examples of ordinal data.

Check Your Progress 1

Define quantitative data. Describe the various types of quantitative data along with examples.

Notes: (a) Space is given below for your answer.
(b) Compare your answer with the one given at the end of the Unit.

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1.3.2 Tabulation and Organisation of Quantitative Data

A. Frequency Distribution

Data collected from a test and by using other gathering/measuring tools are raw and may have little meaning to the researcher until they are tabulated and organised in a systematic order. One of the ways of doing so is to prepare a **frequency distribution**. The method for tabulating the quantified data in a frequency distribution can be illustrated by considering the following scores of 40 students of MA (DE) of the Indira Gandhi National Open University in course MDE-412.

Table 1.1: Tabulation of Scores on a Test in Course MDE-412

57	70	80	82	87
60	72	80	82	88
64	73	80	82	87
67	70	78	80	93
67	76	77	84	95
62	76	78	85	97
61	75	80	85	98
63	70	78	85	90

It is difficult to see from the above table how the scores are distributed. Inspection of scores, however, shows that many scores occur more than once.

You will observe that there are one 98, one 97, one 95, one 88, two 87s, three 85s, and so on. For our convenience, you can arrange the data in columns as shown in Table 1.2. In one column, you can arrange the marks in class-intervals and in the other, you can record the number of students who have scored these marks by tallies. Inspection of the scores in Table 1.1 shows that the highest score is 98 and the lowest is 57. The range is 41 (i.e. 98-57). Therefore, the distribution of scores can be conveniently arranged by dividing the range of 41 into eight or more class-intervals if the classes are taken to be of 5 points each. If you take the starting point at 56, the scores within the range 56 to 60, that is all scores with the values 57 and 60 will be grouped together to form the lowest class-interval. All scores from 61 to 65, that is, 61, 62, 63, 64 and 65 will form the next class-interval. Similarly you shall group all scores within the ranges 66 to 70, 71 to 75 and so on. The highest class interval will be 96-100.

Table 1.2: Frequency Distribution of the Scores of 40 Students: Course MDE-412

Class Interval	Tallies	Frequency (f)
96 -100	II	2
91 - 95	II	2
86 - 90	IIII	4
81 - 85	~~IIII~~ II	7
76 - 80	~~IIII~~ ~~IIII~~ I	11
71 - 75	III	3
66 - 70	~~IIII~~	5
61 - 65	IIII	4
56 - 60	II	2
	Total number of scores N = 40	

In Table 1.2, the class-intervals have been arranged serially from the smallest at the bottom to the largest at the top, each class-interval covering 5 scores. For each score, we have marked a 'tally' against the corresponding class-interval. The first score, 57, is represented by a tally placed against the class interval 56-60, the second score of 60 by a 'tally' marked against the class interval 56-60, and the third score 64 by a tally against the class interval 61-65. The remaining scores have been tabulated in the same way. When all the 40 scores are listed, the total number of tallies in each class-interval are counted and written in the next column f. The total of 'f' gives the total number of scores (in the present case 40) and is denoted by N.

It may be noted that the interval 56-60 takes care of all the scores from 56 upto 60. The score of 56 ordinarily means the interval 55.5 to 56.5 and that the score of 60 means 59.5 to 60.5. The mid-point of the bottom most class-interval is 58. Hence, the distribution represented in Table 1.2 may also be expressed as:

Table 1.3: Frequency Distribution of the Scores of 40 Students: Course MDE-412

Score Intervals	Exact Units of Class-Intervals	Mid Point (x)	(f)
96-100	95.5-100.5	98	2
91-95	90.5-95.5	93	2
86-90	85.5-90.5	88	4
81-85	80.5-85.5	83	7
76-80	75.5-80.5	78	11
71-75	70.5-75.5	73	3
66-70	65.5-70.5	68	5
61-65	60.5-65.5	63	4
56-60	55.5-60.5	58	2
			N = 40

Check Your Progress 2

Tabulate the following data in a frequency distribution using an interval of 5 units.

185	176	166	177	171
147	176	170	171	180
173	168	181	165	173
175	158	156	162	173
197	151	153	162	188
166	145	191	164	174
178	142	158	167	178
148	187	172	169	184
156	187	172	193	183
181	161	172	179	179

Notes: (a) Space is given below for writing your answer.

(b) See whether you have followed the procedure as shown in Tables 1.2 and 1.3.

(c) The answer is incorporated in Check Your Progress 3.

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B. Cumulative Frequency Distribution

In some cases, you may not be concerned with the frequencies within the class-intervals, but rather with the number or the percentage of values greater than or less than a

specified value. These values, called 'cumulative frequencies' or 'cumulative percentage frequencies' are obtained by adding successively the individual frequencies of class-intervals.

Table 1.4: Cumulative Frequency Distribution of the Test Scores of 40 Students: Course MDE - 412

Score Intervals	Exact Units of Class-intervals	Frequency (f)	Cumulative Frequency (F)	Cumulative Percentage Frequency
96-100	95.5-100.5	2	38+ 2 = 40	100.00
91-95	90.5-95.5	2	36+ 2 = 38	95.00
86-90	85.5-90.5	4	32+ 4 = 36	90.00
81-85	80.5-85.5	7	25+ 7 = 32	80.00
76-80	75.5-80.5	11	14+11 = 25	62.50
71-75	70.5-75.5	3	11+ 3 = 14	35.00
66-70	65.5-70.5	5	6 + 5 = 11	27.50
61-65	60.5-65.5	4	2 + 4 = 6	15.00
56-60	55.5-60.5	2	2	5.00
		N = 40		

1.3.3 Graphical Presentation of Quantitative Data

Graphical presentation often facilitates understanding of a set of data. With the help of a well-drawn graph, the data can be read and interpreted very easily. Brief descriptions of the various types of graph which are useful in visualizing the important properties of a frequency distribution are given below.

The following three types of graph are commonly used for the above mentioned purposes:

i) Histogram or column diagram
ii) Frequency polygon
iii) Cumulative percentage curve or ogive.

i) Histogram or column diagram

A **histogram** or **column diagram** is a graph in which class-intervals are represented along the horizontal axis and their corresponding frequencies are represented by areas in the form of rectangular vertical bars drawn on the intervals.

The following steps are followed in preparing a histogram:

Step 1: A horizontal line is drawn at the bottom of a graph paper. Units representing class-intervals are marked along this line.

Step 2: A vertical line is drawn at the left hand extreme of the horizontal axis. Along this vertical axis, units representing individual frequencies of the class-intervals are marked.

Step 3: Taking class units as bases, rectangles are drawn, such that the areas of rectangles are proportional to the frequencies of the corresponding classes.

Let us consider the following data for drawing a histogram as an illustration of what you have read above.

Table 1.5: Frequency Distribution of the Scores of 40 Students: Course MDE-412

Class Intervals	Exact Units of Class Intervals	Mid Point	Frequency (f)	Cummulative Frequency (F)	Cummulative Percentage Frequency
35 - 39	34.5 - 39.5	37	4	40	100.00
30 - 34	29.5 - 34.5	32	8	36	90.00
25 - 29	24.5 - 29.5	27	11	28	70.00
20 - 24	19.5 - 24.5	22	8	17	42.50
15 - 19	14.5 - 19.5	17	6	9	2.50
10 - 14	9.5 -14.5	12	3	3	7.50
			N = 40		

The histogram drawn for the above data is shown in figure 1.

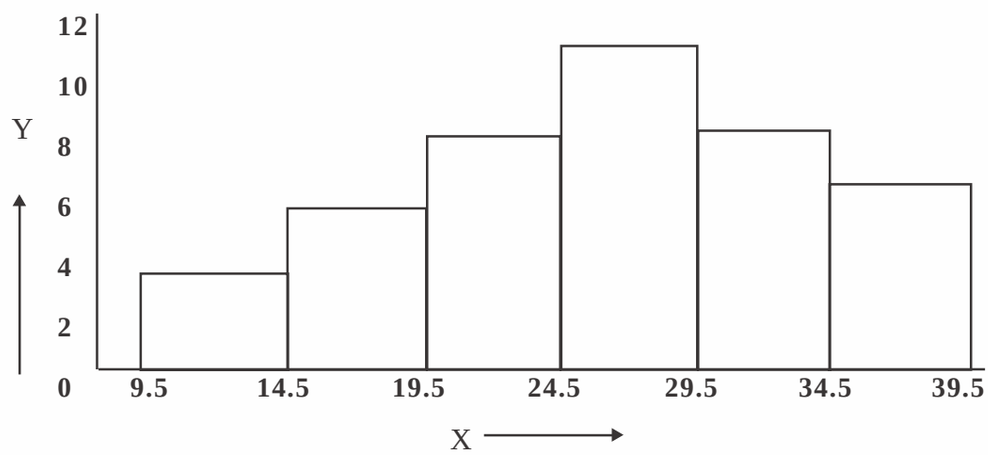


Fig. 1: Histogram plotted from the data of Table 1.5

ii) Frequency Polygon

Frequency polygon is drawn by plotting the mid-point of each class-interval at a height proportional to its respective frequency and then joining the points by straight lines. The first two steps are identical to those used in the construction of a histogram. The next step to be followed is given as under:

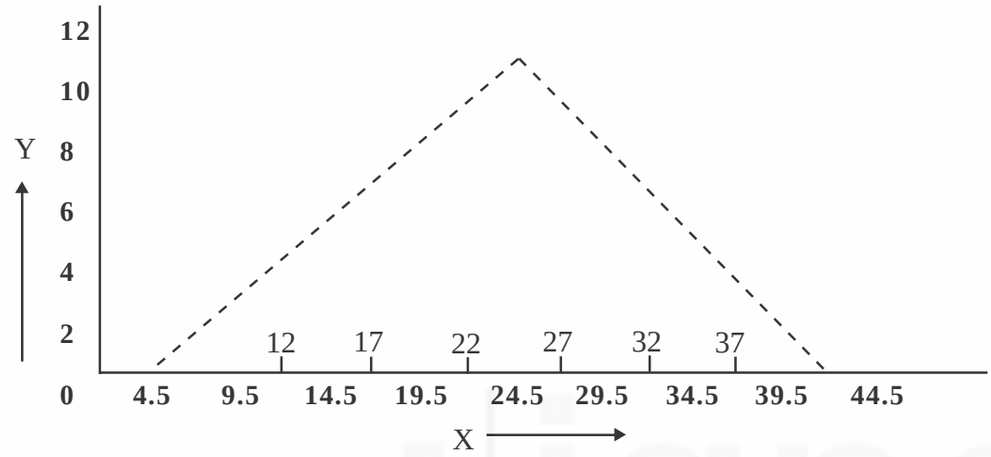


Fig. 2: Frequency Ploygon plotted from the data of Table 1.5

Step 3: Directly above the mid-point of each class-interval along the horizontal axis plot the points at a height proportional to the respective frequencies. Join these points by straight lines. The frequency polygon for the distribution of table 1.5 is shown in the figure 2.

iii) Cumulative Percentage Curve or Ogive

When the frequencies are expressed as **cumulative percentages** of N on the vertical axis, the graphic representation is known as a cumulative percentage curve or ogive. After finding the cumulative percentage frequencies, the points are plotted on the exact upper limits of the class-intervals. A curve joining the points thus obtained is called the **cumulative percentage curve** or **ogive**.

The cumulative percentage curve or ogive of the distribution represented in table 1.5 is illustrated in figure 3.

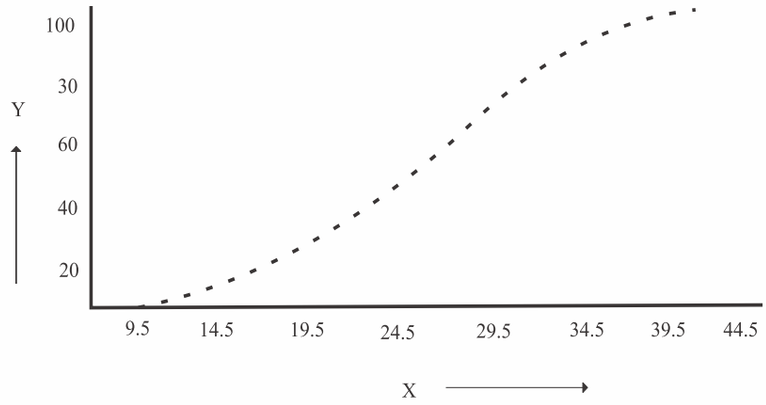


Fig. 3: Cumulative percentage curve or ogive plotted from the data of Table 1.5.

1.3.4 Analysis of Quantitative Data

Analysis of quantified data means studying the organised or tabulated data in order to discover the inherent facts. The data are studied from as many angles as possible to explore the new facts. Two types of statistical methods are used in the analysis of the tabulated data measured/expressed in quantified terms. The first category of methods pertain to **'descriptive analysis'** and the second, to **'inferential analysis'** of data.

In this Unit, you will be concerned with 'descriptive analysis'. Analysis of quantitative data can be also done by using computer software like SPSS, SAS, Stata and XL Stat. If you are interested to know more about these software you can check the websites. But it require thorough understanding of the program and computer.

Descriptive statistical analysis limits generalisation to the particular observed group of individuals. This analysis describes only one single group. The computed statistical values provide valuable information about the nature of that particular group only.

The following methods are generally used in descriptive statistical analysis of the tabulated data:

i) **Measures of central tendency**
ii) **Measures of variability**
iii) **Measures of relative positions**
iv) **Measures of relationships.**

We shall touch upon each one of them in some detail as follows:

A. Measures of Central Tendency

The three most commonly used measures of central tendency are the **Mean**, the **Median** and the **Mode**.

I) The Mean (M)

The arithmetic average of a distribution is known as its mean. The mean of a set of observations or measures is obtained by dividing the sum of all values by the total number of values.

a) Mean for an ungrouped data:

The formula for finding the mean for an ungrouped data is:

$$M = \frac{\sum X}{N} \qquad \text{.....(1)}$$

in which

M = Mean

$\sum$ = Sum of

X = Observations in a distribution
N = Total number of observations.

To illustrate the use of formula (1) let us consider the data:

16, 14, 12, 18, 21, 22, 13, 15, 16, 18

Using the formula (1):

$$M = \frac{16 + 14 + 12 + 18 + 21 + 22 + 13 + 15 + 16 + 18}{10}$$

$$= \frac{\sum X}{N} = 16.5.00$$

b) Mean for grouped data

When the number of observations or measures is large, the data is grouped in a frequency distribution.

The mean is computed by the formula:

$$M = AM + \frac{\sum fx'}{N} \times i \qquad \text{.....(2)}$$

Where

M = Mean
AM = Assumed Mean
x' = [Midpoint score(x) - AM]/(length of the class interval)
$\sum fx'$ = Sum of the products of frequencies and deviation of observations from the assumed mean.
i = Width of the class-interval
N = Total number of observation

To illustrate the use of formula (2), consider the grouped data given in table 1.5.

Computations

Step 1: Put the class-intervals in exact limits
Step 2: Find the mid-point of each class interval and take the assumed mean at the interval which has the maximum frequency.
Step 3: Find the difference between each mid-point score and the assumed mean and divide it by the length of the class-interval to get the deviation x.
Step 4: Compute fx for each class-interval (fx is the product of the frequency and deviation of the observation from the assumed mean in a particular case.)

Step 5: Find the sum of all fx
Step 6: Apply formula (2) to compute the mean.

Table 1.6: The Calculation of the Mean from Data Grouped into a Frequency Distribution (Ref. Table)

Class Interval	Mid-point (x)	Frequency *(f)*	Deviation from the AM x	fx
34.5 - 39.5	37	4	2*	8
29.5 - 34.5	32	8	1	8
24.5 - 29.5	**27 AM**		11	0 0 (+16)
9.5 - 24.5	22	8	-1	-8
14.5 - 19.5	17	6	-2	-12
9.5 - 14.5	12	3	-3	-9 (-29)
		N = 40		

$$*2 = \frac{37 - 27}{5}$$

Using the formula (2):

$$M = AM + \frac{\sum fx'}{N} \times i$$

$$= 27.0 + \frac{-13}{40} \times 5$$

$$= 27.0 - 1.625$$

$$= 25.375$$

II) The Median

The median is a point in an array, above and below which one half of the observations fall. It is a measure of position rather than magnitude.

a) Median for Ungrouped data

If the observations are ungrouped and their number is small, the observations are arranged in the order of magnitude. The middle score is determined by counting up half the value of N if the number of observation (N) is even. When the number of observations (N) is odd, the mid-observation value is median. For example, 10 is the median of scores : 7, 8, 9, 10, 11, 12, 13. When the number of scores (N) is even, the

median is the mid-point between the two middle scores. For example:

$\frac{(10+11)}{2} = 10.5$ is the median of scores:

7, 8, 9, 10, 11, 12, 13, 14.

b) Median for grouped data

In the case of grouped data, cumulative frequency distribution is prepared and the median is calculated by the formula:

$$Mdn = l + \frac{N/2 - F}{f} \times i \qquad .....(3)$$

Mdn = Median
l = Exact lower limit of the class-interval upon which the median lies.
N/2 = One half of the total number of observations
F = Sum of all frequencies below l.
f = Frequency within the class-interval upon which the median lies.
i = Width of the class interval in which the median lies.

To illustrate the use of formula (3) consider the data of table 1.5 once again.

Table 1.7: The Calculation of the Median from Data Grouped into a Frequency Distribution

	Class-Interval	Frequency (F)	Cumulative Frequency (F)
	34.5 – 39.5	4	40
	29.5 – 34.5	8	36
Median Class	24.5 – 29.5	11	28
	19.5 – 24.5	8	17
	14.5 – 19.5	6	9
	9.5 – 14.5	3	3
		N = 40	

Here N/2 = 20, I = 24.5, F = 17, f =11 and i = 5

Using formula (3)

$$\text{Mdn} = l + \frac{N/2 - F}{f} \times i$$

$$= 24.5 + \frac{(20 - 17)}{11} \times 5$$

$$= 24.5 + \frac{15}{11}$$

$$= 25.86$$

III) The Mode

The mode is defined as the most frequently occurring measure of an observation in a distribution. If only one value occurs a maximum number of times the distribution is said to have one mode; i.e. the distribution is unimodal. In some distributions there may be more than one mode. A two mode distribution is bimodal and it is multimodal in a distribution, which has more than two modes.

a) Mode for Ungrouped Data

In a simple ungrouped series of measures, the crude or empirical mode is that single measure which occurs most frequently. For example, in the series 7, 8, 9, 10, 11, and 12 the most often recurring measure, namely, 9 is the crude or empirical mode.

b) Mode for grouped data

When data are grouped into a frequency distribution, the crude or empirical mode is usually taken to be the mid-point of that interval which contains the largest frequency. In the example given in table 1.5, the interval 24-29 contains the largest frequency and hence 26.5, its mid-point, is the crude mode.

The true mode, that is, the point of greatest concentration in the distribution, or the point at which more measures fall than at any other point, is calculated by the formula:

$$\text{Mode} = l + \frac{fm - f_1}{2fm - f_1 - f_2} \times i \qquad \text{.....(4)}$$

Where

l	=	Lower limit of the modal class i.e., the class interval having maximum frequency
fm	=	Frequency of the modal class.
f_1	=	Frequency of the class-interval preceding the modal class.
f_2	=	Frequency of the class-interval following the modal class.
i	=	Width of the modal class.

To illustrate, let us make use of formula (4) for the data in table 1.5. Here the maximum frequency is 11 which lies in class interval 24.5 - 29.5.

Therefore, the modal class is (24.5 - 29.5).

Here *fm* = 11, 8 1 = f , 8 2 = f , i =5 and l = 24.5.

Using Formula (4) :

$$\text{Mode} = l + \frac{fm - f_1}{2fm - f_1 - f_2} \times i$$

$$= 24.5 + \frac{11 - 8}{2 \times 11 - 8 - 8} \times 5$$

$$= 24.5 + \frac{3}{6} \times 5$$

$$= 24.5 + 2.5$$

$$= 27.00$$

B. Measures of Variability

The measures of central tendency are very useful in describing the nature of a distribution of measures, but they do not give the researcher a complete picture of the data. These measures will not tell the researcher how the scores tend to be distributed. For this, you use a different set of measures which are called measures of **'variability'** or measures of '**spread**' or '**dispersion'**. The most commonly used measures of variability include the **range**, the **variance** and **standard deviation**.

I. The Range

The range is defined as the difference between the two extreme measures or values in a distribution. Suppose the scores of 10 learners in the course MDE -412 are :

50, 40, 39, 35, 29, 28, 24, 27, 19. 18.

The range for this distribution will be (50-18) = 32. Although the range has the advantage of being easily calculated, it has the following serious limitations:

1) As the value of range is based on only two extreme values in the total distribution, it does not give any idea of the variation of many other values of the distribution.

2) It is not a stable statistic as its value can differ from sample to sample drawn from the same population.

II. The Variance and Standard Deviation

The average of the squared deviations of the measures or values from their mean is known as the variance. The standard deviation is the positive square root of variance.

a) The Variance and Standard Deviation for the Ungrouped Data

The variance for the ungrouped data is found by using the formula:

$$\sigma^2 = \frac{\sum x^2}{N} \quad \text{.....(5)}$$

= Variance of the sample

x = Deviation of the raw measures or values from the mean.

N = Number of values or measures

Let us consider the following data of scores for the application of formula (5):

10, 10, 9, 9, 8, 8, 7, 7, 6, 6.

As the deviation of each score from the mean is required, the first thing to do is to calculate the mean. Using formula (1)

$$M = \frac{\sum x}{N} = \frac{80}{10} = 8$$

Now, from each raw score, the mean is substracted to get the value of x.

Table 1.8: Distribution of the Test Scores of Ten Learners of Course MDE - 412

Score (X)	Deviation (X-M) (x)	Deviation Squared (x^2)
10	2	4
10	2	4
9	1	1
9	1	1
8	0	0
8	0	0
7	-1	1
7	-1	1
6	-2	4
6	-2	4
		----x^2 =20

Using formula (5)

$$\sigma^2 = \frac{\sum x^2}{N}$$

$$= \frac{20}{10} = 2$$

Now to get the standard deviation, you need the positive square root of the variance, σ^2.

Standard Deviation = $\sigma = \sqrt{\frac{\sum x^2}{\mathrm{N}}}$

$$= \sqrt{2}$$

$$= 1.41$$

The raw scores instead of deviation scores may also be used. The raw score formulae for variance and standard deviation are given as follows

Variance $= \sigma^2 = \frac{N\sum X^2 - (\sum X)^2}{N^2}$ .....(6)

Standard Deviation $= \sigma = \frac{\sqrt{N\sum X^2 - (\sum X)^2}}{N^2}$ .....(7)

In which

X = Raw score

N = The number of scores in the distribution.

Using the same set of data, you can calculate variance and standard deviation with the help of raw-score formulae:

Table 1.9: The calculation of variance and standard deviation from original (row) scores when the assumed mean is taken at zero and the data is ungrouped

Score (X)	X^2
10	100
10	100
9	81
9	81
8	64
8	64
7	49
7	49
6	36
6	36
....X =80	.....X^2 = 660

Using formula (6)

$$\text{Variance} = \frac{N\sum X^2 - (\sum X)^2}{N^2}$$

$$= \frac{10 \times 660 - (80)^2}{100}$$

$$= \frac{6600 - 6400}{100}$$

$$= 2$$

Using formula (7)

$$\text{Standard Deviation} = \sigma = \frac{\sqrt{N\sum X^2 - (\sum X)^2}}{N}$$

$$= \sigma = \frac{\sqrt{10 \times 660 - (80)^2}}{10}$$

$$= \frac{\sqrt{6600 - 6400}}{10}$$

$$= \frac{14.14}{10}$$

$$= 1.414$$

b) Variance and Standard Deviation for Grouped Data

In the case of grouped data in a frequency distribution, the variance and standard deviation are calculated by using the formulae:

$$\text{Variance} = = \sigma^2 = \frac{i^2}{N^2}\left[N\sum fx'^2 - \left(\sum fx'^2\right)^2\right] \quad \text{.....(8)}$$

$$\text{Standard Deviation} = \frac{i^2}{N^2}\left[\sqrt{N\sum fx'^2 - \left(\sum fx'^2\right)^2}\right] \quad \text{.....(9)}$$

Where

i = Width of the class-interval
N = Total number of measures
f = Frequency of class-interval
x^1 = Deviation of the raw measure from the assumed mean divided by the length of class-interval.

To illustrate the use of these formulae let us consider the distribution given in table 1.10.

Table 1.10: The Calculation of Variance and Standard Deviation from Data Grouped in a Frequency Distribution

Class Interval	x	f	x^1	fx^1	fx^{12}
71-75	73	3	3	9	27
66-70	68	4	2	8	16
61-65	63	9	1	9	9
56-60	58	15	0	0	0
51-55	53	8	-1	-8	8
46-50	48	6	-2	-12	24
41-45	43	5	-3	-15	45
		N = 50		…….fx1 =-9	…..fx12 =129

Using formula (8)

$$\text{Variance} = \sigma^2 = \frac{i^2}{N^2}\left[N\sum fx'^2 - \left(\sum fx'^2\right)^2\right]$$

$$= \frac{(5)^2}{(50)^2}\left[50\times129 - (-9)^2\right]$$

$$= 63.69$$

Using formula (9)

Standard Deviation $= \frac{i^2}{N^2}\left[\sqrt{N\sum fx'^2 - \left(\sum fx'^2\right)^2}\right]$

$$= \frac{5}{50}\sqrt{\left(50\times 129 - (-9)^2\right)}$$

$$= \frac{1}{10}\sqrt{6369}$$

$$= \frac{1}{10}\times 79.81$$

$$= 7.98$$

The standard deviation is a very useful device for comparing characteristics that may be different or expressed in different units of measurement. It is also used in describing the status or position of an individual in a group. But before this concept is developed further, it is essential to understand the nature of the 'normal probability distribution'.

Check Your Progress 3

Compute (i) Mean (ii) Variance and (iii) Standard Deviation for the following frequency distribution:

Class Interval	F
195-199	1
190-194	2
185-189	4
180-184	5
175-179	8
170-174	10
165-169	6
160-164	4
155-159	4
150-154	2
145-149	3
140-144	1

Notes: (a) Space is given below to write your answer.
(b) Compare your answer with the one given at the end of this unit.

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Normal Probability Distribution

The **normal probability distribution** is based upon the law of probability. It is not an actual distribution of measures or scores; instead, it is a mathematical model. It is represented by a curve which is called the Normal Probability Curve. Figure 4 represents an ideal normal probability curve.

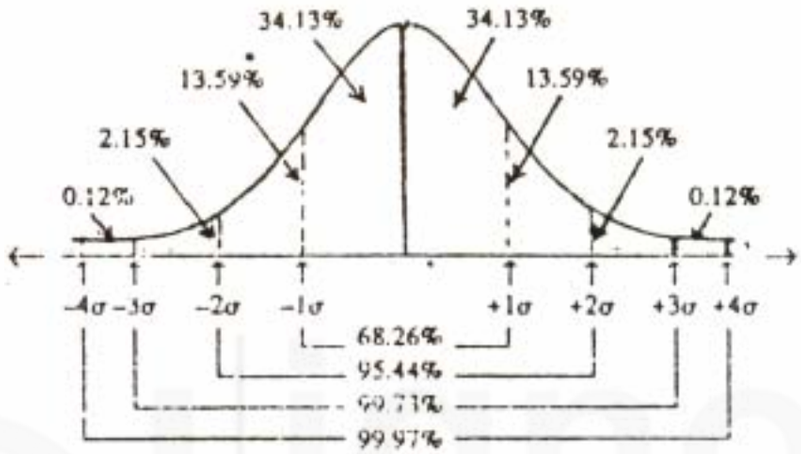


Fig. 4: Ideal Normal Probability Curve

The normal probability curve has the following characteristics:

1. The curve is symmetrical around its vertical axis called ordinate. It implies that the size, shape and slope of the curve on one side of the ordinate is identical to that on its other side.
2. The values of mean, mode and median computed for a distribution following this curve, are always the same.
3. The height of the vertical line called ordinate is maximum at mean and in the unit normal curve it is equal to 0.3989.
4. The curve is asymptotic. It approaches but does not meet horizontal axis and extends from $-\infty$ (minus infinity) to $+\infty$ (plus infinity).
5. The points of inflection of the curve occur at points ± 1, standard deviation $(\pm 1\sigma)$, above and below the mean. Thus the curve changes from convex to concave in relation to the horizontal axis at these points.
6. About 68.26 percent of the total area falls between the limits M + 1σ and M –1σ ; 95.44 percent of the total area of the curve falls between the limits M + 2σ and M - 2σ and 99.73 percent of the total area of the curve falls between M+ 3σ and M - 3σ .

However, these calculations are rarely necessary, as Normal Table is available from which the information about the area is readily available. For this reason it is very essential that the use of Normal Table (Table I Appendix, Unit 2) be clearly understood. Table I gives the fractional parts of the total area under the normal curve found between

mean and ordinate (Y's) erected at various distances from the mean. The total area under the curve is taken arbitrarily to be 10,000, because of the greater convenience with which fractional parts of the total area may then be calculated. You know that x = (X-M) measures the deviation of a raw score (X) from the mean (M). If x is divided by σ, then this deviation is expressed in σ units. These σ deviation scores are called sigma scores or z-scores $\left(i.e. z = \frac{X - M}{\sigma} = \frac{x}{\sigma} \right)$. The first column of the table under $\frac{x}{\sigma}$ gives distance from the mean in the tenth of σ and distance from the mean in the hundredth of σ are given by the headings of the other columns.

To find the number of cases in the normal distribution between the mean and the ordinate erected at a distance of 1σ from the mean, you go down the $\frac{x}{\sigma}$ column until 1.0 is reached, and in the next column under .00 you take the entry opposite to 1.0, namely 34.13. This figure means that 3413 cases in 10,000, or 34.13 percent of the entire area of the curve lies between the mean and 1σ. Similarly, if you have to find out the percentage of the distribution between mean and 1.65σ, you go down the $\frac{x}{\sigma}$ column till 1.6, then across horizontally to the column headed .05, and take the entry 45.5.

This shows that in a normal distribution, 45.05 percent of the total area lies between the mean and 1.65σ.

Check Your Progress 4

Describe the characteristics of a Normal Probability Distribution

Notes: (a) Space is given below for writing your answer.
(b) Compare your answer with the one given at the end of this unit.

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C. Measures of Relative Positions

A raw score on a test, taken by itself, has no meaning. It gets meaning only by comparison with some reference group or groups. for eg. If a student score 50 in Maths and 30 in Science, it does not mean that the student did better in Maths. 50 may be the lowest score in Maths test and 30 may be the highest score in Science test. The comparison may be done with the help of the following measures:

1. **Sigma Scores**
2. **Standard Scores**
3. **Percentiles**
4. **Percentile Ranks.**

What does each one of these mean?

1. Sigma Scores

A **sigma score** makes a realistic comparison of scores possible and provides a basis for equal weighting of the scores as the scores on different tests are expressed on a scale with a mean of zero and standard deviation of 1.

Let us suppose that the mean of a test is 75 and the standard deviation is 5.0. Then if A earns a score of 85 on this test, his/her deviation from the mean is 85-75 = 10. Dividing this deviation of 10 by the standard deviation (σ), i.e., 5.0, we give him a score of $\frac{10}{5} = 2$. If B's score on this test is 64, his/her deviation from the mean is 64-75 = -11 and the score in σ units is -2.20. Deviations from the mean expressed in σ terms are called sigma scores.

Half of the scores in a distribution lie below and half above the mean, about half of σ scores are positive and half are negative.

2. Standard Scores

The sigma scores, which are often small decimal fractions and half of them are negative, are somewhat inconvenient to deal with. Hence, scores are usually converted into a new distribution with mean and standard deviation so selected that it makes all scores positive and relatively easy to handle in computation. Such scores are called **'standard scores'.**

The formula for the conversion of a raw score to a standard score is as follows:

$$X' = \frac{\sigma'}{\sigma}(X - M) + M' \qquad \text{.....(10)}$$

in which

X' = A standard score in a new distribution

σ' and σ = SD's of standard and raw scores.

X = A score in the original distribution.

M and M = Means of raw and standard scores.

When the mean (M') and standard deviation (σ') are taken to be 50 and 10 respectively, the standard score is called a T-score.

i.e. $T = \frac{10}{\sigma}(X - M) + 50$ .....(11)

Example: To illustrate, let us consider a distribution with its mean 67 and σ = 12.5. Let us also suppose that A's score is 76 and B's score is 54.

Express these scores as (i) standard scores in a distribution with a mean of 250 and σ of 50 and (ii) T-scores.

Using formula (10)

$$X' = \frac{50}{12.5}(X - 67) + 250$$

Substituting A's score of 76 in the above equation you have:

$$X' = \frac{50}{12.5}(76 - 67) + 250$$

$$= \frac{50 \times 9}{12.5} + 250$$

$$= 286$$

Substituting B's scores of 54 in the above equation

$$X' = \frac{50}{12.5}(54 - 67) + 250$$

$$= 198$$

Using formula (11)

$$T = \frac{10}{12.5}(X - 67) + 50$$

Substituting A's score in the above equation you have:

$$T = \frac{10}{12.5}(76 - 67) + 50$$

$$T = 0.8 \times 9 + 50$$

$$= 57.2$$

Substituting B's score in the above equation you have:

$$T = \frac{10}{12.5}(54 - 67) + 50$$

$$= 39.6$$

3. Percentiles

Percentiles are the points which divide the entire scale of measurement into 100 equal parts. They are denoted by P_0, P_1, P_2, P_3, P_4, P_5 …… P_{99}, and P_{100}.



The first percentile may be defined as that point in a frequency distribution below which lie 1 percent of the total measures or scores. Similarly, twentieth percentile may be defined as that point in a frequency distribution below which 20 percent of the total measures or scores fall. It is evident that the median, expressed as a percentile, is P_{50}. It should be noted that P_0 lies at the beginning of the distribution and P_{100} at the end of the distribution.

The formula for calculating percentiles is

$$P_p = l + \frac{(PN - F)}{f_p} \times i \qquad \text{.....(12)}$$

in which

P_p = percentile of the distribution wanted

l = exact lower limit of class-interval upon which P_p lies.

PN = part of the N to be counted off in order to reach P_p.

F = sum of all scores upon intervals below l.

fp = number of scores within the interval upon which P_p falls

i = length of the class-interval.

The use of formula (12) may be illustrated by the following example.
Calculate P_{25}, P_{45} and P_{95} from the following distribution:

Table 1.11: The Calculation of Percentiles from Data Grouped in a Frequency Distribution

Scores Class-intervals	Frequency (f)	Cumulative Frequency (F)
81.5 - 86.5	1	8
76.5 - 81.5	4	79
71.5 - 76.5	5	75
66.5 - 71.5	10	70
61.5 - 66.5	35	60
56.5 - 61.5	12	25
51.5 - 56.5	9	13
46.5 - 51.5	2	4
41.5 - 46.5	2	2
	N=80	

For computing P_{25} you have to first find PN

Here, 25 percent of 80 is 20, PN = 20

Now l = 56.5, F = 13, fp = 12 and i = 5

Using formula (12)

$$P_{25} = 56.5 + \frac{20-13}{12} \times 5$$

$$= 56.5 + 2.92 = 59.42$$

Similarly

$$P_{45} = 61.5 + \frac{36-25}{35} \times 5$$

$$= 61.5 + 1.57 = 63.07$$

$$P_{95} = 76.5 + \frac{76-75}{4} \times 5$$

$$= 76.5 + 1.25 = 77.25$$

4. Percentile Ranks

The **percentile rank** is the point in the distribution below which a given percentage of scores falls. If the 80th percentile rank is a score of 65, then 80 percent of the scores falls below 65. The median is the 50th percentile rank, for, 50 percent of the scores fall below it.

The process of calculating percentile ranks is the reverse process of calculating percentile points. You have to calculate ranks corresponding to particular scores. If R is the rank and N is the total number of cases, then:

$$\text{Percentile Rank} = 100 - \frac{100R - 50}{N} \quad \text{..... (13)}$$

Suppose A ranks 13th in the class of 80 learners, 12 learners rank above it, 67 below it. Its percentile rank is :

$$= 100 - \frac{100 \times 13 - 50}{80}$$

$$= 100 - 15.625$$

$$= 84$$

D. Measures of Relationship

The data in which we secure measures of two variables for each individual is called **bivariate data**. The essential feature of bivariate data is that one measure can be

compared with another measure for each member of the group. When bivariate data are studied, you may like to know the degree of relationship between the variables of such data. This degree of relationship is known as correlation. It can be quantitatively represented by the **coefficient of correlation**. Its value ranges from -1.00 to +1.00. A value of -1.00 describes a perfect negative correlation and +1.00 describes perfect positive correlation. A zero value describes complete lack of correlation between the two variables. The sign of the co-efficient indicates the direction of relationship and numerical value is its strength/magnitude.

Methods of correlating variables

There are various methods of correlating variables. Their use is relative to the situation and type of data. **Product Moment Correlation** and **Rank Order Correlation** are mostly used for computing correlation between two variables.

1. Product-moment correlation

In some situations, the data for two variables X and Y are expressed in interval or ratio level of measurement and the distributions of these variables have a linear relationship. Moreover, the distributions of variables are uni-modal and their variances are approximately equal. In such situations, product moment method of correlation is used generally. It is also called **Pearson's r**.

i) Calculation of Pearson's r from ungrouped data:

When the size of the sample is small, there is no need of grouping the data and Pearson's r may be calculated with the help of the following formula:

$$r_{xy} = \frac{N\sum xy - \left(\sum x\right)\left(\sum y\right)}{\sqrt{\left[N\sum x - (x)^2\right]\left[N\sum y^2 - \left(\sum y\right)^2\right]}} \qquad \text{.....(14)}$$

in which

x = deviations of X measures from the assumed mean.
y = deviations of Y measures from the assumed mean.

To illustrate the use of formula (14), let us compute the product moment 'r' from the following data for the two variables X and Y for 10 learners who are enrolled in a Study Centre of an Open University.

X : 45 54 52 58 62 46 55 49 50 54

Y : 42 50 55 46 59 41 46 48 45 48

Using formula (14) for the data in Table 1.12

Table 1.12: The Calculation of Product Moment Correlation from Ungrouped Data when Deviations are taken from Assumed Mean

X	Y	x	y	x^2	y^2	xy
45	42	-7	-6	49	36	42
54	50	2	2	4	4	4
52(AM)	55	0	7	0	49	0
59	46	6	-2	36	4	-12
62	59	10	11	100	121	110
46	41	-6	-7	36	49	42
55	46	3	-2	9	4	-6
49	48(AM)	-3	0	9	0	0
50	45	-2	-3	4	9	6
54	48	2	0	4	0	0
		$\sum x = 5$	$\sum y = 0$	$\sum x^2 = 251$	$\sum y^2 = 276$	$\sum xy = 186$

Using formula (14)

$$r_{xy} = \frac{N\sum xy - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2[(N\sum y^2 - (\sum y)^2]}}$$

$$= \frac{10 \times 186 - 5 \times 0}{\sqrt{[10 \times 251) - (5)^2][10 \times 276 - (0)^2]}}$$

$$= \frac{1860}{2618.89}$$

$$= 0.71$$

ii) Calculation of Pearson's r from grouped data:

When N is large or even moderate in size, and when no calculating machine is available, the best procedure is to group data in both variables X and Y and to form a scattergram.

The values from the scattergram may be used in the following formula:

$$r_{xy} = \frac{N\Sigma fxy - (\Sigma fx)(\Sigma fy)}{\sqrt{[N\Sigma fx^2 - (\Sigma fx)^2][(N\Sigma fy^2 - (\Sigma fy)^2]}} \qquad \text{.....(15)}$$

To illustrate the use of the formula (15) consider the data of 50 learners enrolled with IGNOU in Course X and in Course Y in the following scattergram:

COURSE (Y)

COURSE X

Y \ X	20-21	22-23	24-25	26-27	28-29	30-31	32-33	$f_{(y)}$	y	fy	fy^2	fxy
51-53			$_{-2}1^{-2}$	$_{0}2^{0}$				3	2	6	12	-2
48-50	$_{-6}2^{-3}$	$_{-4}2^{-2}$	$_{-3}3^{-1}$	$_{0}3^{0}$				10	1	10	10	-13
45-47	$_{0}1^{0}$	$_{0}3^{0}$	$_{0}4^{0}$	$_{0}8^{0}$	$_{0}3^{0}$	$_{0}2^{0}$	$_{0}1^{0}$	22	0	0	0	0
42-44	$_{2}2^{3}$	$_{4}2^{2}$	$_{3}3^{1}$	$_{0}5^{0}$				12	-1	-12	12	13
39-41		$_{4}1^{4}$	$_{2}1^{2}$	$_{0}1^{0}$				3	-2	-6	12	6
$f_{(x)}$	5	8	12	19				50		-2	46	04
x	-3	-2	-1	0	1	2	3					
fx	-15	-16	-12	0	3	4	3	-33				
fx^2	45	32	12	0	3	8	9	109				
fxy	0	4	0	0	0	0	0	04				

Fig. 5: A Scattergram Showing Paired Scores of 50 Learners on the Tests of Course X and Course Y.

The computation for the values Σfxy, Σfx^2, Σfxy etc., may be done in the following steps in the order given below.

Step 1

The distribution of Course X scores for the 50 learners is found in the f(y) column on the right of the scattergram. Assume a mean for the distribution of scores of course X (the mid-point of that interval which contains the largest frequency), and draw double lines to mark off the row in which the assumed mean falls. In the present example, the mean

score for course X has been taken at 46 (mid point of interval 45-47) and y's (deviations from the assumed mean) have been taken from this point.

Fill in *fy* and then fy^2 columns.

Step 2

The distribution of the Course Y of 50 learners is found in the f(x) row at the bottom of the scattergram. Assume a mean for this distribution and draw double lines to designate the column under the assumed mean. The mean for the Course Y scores is taken at 26.5 (mid-point of interval 26-27), and the x's (deviations from assumed mean) are taken from this point. Fill in the *fx* and then fx^2 rows.

Step 3

The fxy for a cell is computed by multiplying the frequency given in the particular cell with the corresponding x and y. For example, there is a frequency 1 corresponding with Course Y score 24-25 and Course X score 51-53. The corresponding x for this cell frequency is -1 and corresponding y is +2. Thus fxy for this cell is (1) (-1) (+2) = -2. Similarly the value for fxy is computed for all the cells and their sum is calculated row-wise as well as column-wise. The two sums should equal each other. In the present example, it has come to be 4.

Step 4

Substituting the values for $\sum fx, \sum fx^2, \sum fy, \sum fy^2,$ and $\sum xy$ in the formula (15) we get :

$$r_{xy} = \frac{50\times 4-(-33)(-2)}{\sqrt{\left[50\times 109-(-33)^2\right]\left[50\times 46-(-2)^2\right]}}$$

$$= 0.042$$

2. Rank Order Correlation

It is also known as the **Spearman rank order co-efficient of correlation** and is denoted by **r (rho)**. When the data are available in ordinal (rank) form of measurement rather than in interval or ratio form, this type of correlation is useful.

To find out Spearman rank order coefficient of correlation, the following formula is used:

$$p = 1-\frac{6\sum D^2}{N\left(N^2-1\right)} \qquad \text{.....(16)}$$

in which

D = Difference between the paired ranks.

$\sum D^2$ = Sum of the squared differences between ranks

N = Number of paired ranks.

To make use of formula (16) let us consider the following data. Two judges X and Y ranked 10 distance learners in a declamation contest. The ranks given to them by the judges are given in table 1.13.

Table 1.13: The Calculation of Rank Difference Correlation

Students	Rank assigned by X	Rank assigned by X	D	D2
A	2	3	-1	1
B	4	5	-1	1
C	5	4	1	1
D	10	9	1	1
E	8	7	1	1
F	1	2	-1	1
G	3	1	2	4
H	9	8	1	1
I	6	10	-4	16
J	7	6	1	1
				$\sum D^2$ =28

Using formula (16)

$$p = 1 - \frac{6\sum D^2}{N\left(N^2 - 1\right)}$$

$$= 1 - \frac{6 \times 28}{10\left(100 - 1\right)}$$

$$= 0.83$$

Check Your Progress 5

Compute product moment correlation for the following data:

X : 45, 55, 56, 58, 60, 65, 68, 70, 75, 80, 85

Y : 56, 50, 48, 60, 62, 64, 65, 70, 74, 82, 90

Notes: (a) Space is given below for writing your answer.

(b) Compare your answer with the one given at the end of the Unit.

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1.4 QUALITATIVE DATA

It has already been explained that quantitative measurement makes use of tools that provide a standardised framework in order to limit data collection to certain predetermined responses or categories. The variables that describe a phenomenon are fit in the standardised categories to which numerical values are attached . But in some situations it is difficult to analyse a phenomenon into various components or variables which can be measured in quantified terms. In such cases the researcher takes into consideration the phenomenon as a whole and assumes that there is some quality in the phenomenon in its entirety. When the researcher attempts to retain the totality of a phenomenon while verifying propositions regarding it, he/she adopts a qualitative approach. While using this approach the researcher seeks to capture what people have to say in their own words. **Qualitative approach** describes the experiences of people in depth and permits the researcher to record and understand people in their own perceptions.

Qualitative data consist of 'detailed descriptions' of situations, events, people, interactions, and observed behaviours. These data are also available in the form of 'direct quotations' from people about their experiences, attitudes, beliefs, and thoughts. The verbal data gathered through questionnaires, observations and interviews are mostly qualitative in nature. The 'excerpts' or 'entire passages' from documents, correspondence records and case histories are also examples of data of a qualitative nature. It may be noted that detailed descriptions, direct quotations, and case documentation of a qualitative nature are raw data from empirical situations.

1.4.1 Organisation of Qualitative Data

The responses to open-ended questions on a questionnaire are pretty extensive; they are neither systematic nor standardised. However, they permit the researcher to understand situations as seen and felt by the respondent. The data gathered through participant observation or an open ended/unstructured interview are also descriptive in nature. These descriptions may be in the form of field notes specifying some basic information pertaining to the place where the observation has taken place as well as descriptions about the people who participated in the activities and their extrinsic behaviour in the course of the activities. However, it is not possible to interpret minds while observing their extrinsic behaviour. Through an open-ended/unstructured interview, you can know more about those events which had occurred earlier or could not be observed during participant observation. It provides a framework within which the researcher should be able to gather information from people conveniently and accurately. The information may pertain to a programme, the reaction of participants about the programme and the type of change the participants perceive in themselves after their involvement in the programme. The data are mostly in the form of responses to structured and unstructured questions put to respondents by the researcher during conversation. The responses are generally direct quotations from respondents in their own words and provide details about the situations, events, people, experiences, behaviours, values, customs, etc.

The qualitative data gathered using open-ended questionnaires, participant observations and in-depth interviews are voluminous. They need to be organised and classified into specific patterns, categories and descriptive units to avoid any chaos. However, before any such classification is done, it is advisable to make some copies of the data. One copy may be stored in a safety deposit box so that in case of a loss, this copy can be used by the researcher. The second copy may be used for further treatment of the data throughout. The third copy may be used to fill the missing gaps, identified during scrutiny by the researcher. Additional notes can also be recorded in this copy. Since the organisation of qualitative data involves a lot of cutting and pasting, a fourth copy may be used for that purpose.

Actual classification or organisation of the data can begin only after the copies are made. There are no formal or universal rules for organising the data in various units, patterns, or categories. It requires a creative approach and a lot of perseverance to give a meaningful look to the data. The contents of field notes about interviews or observations may be read carefully by the researcher and he/she may note down his/her comments on the margins

or attach small pieces of paper with his/her written comments/notes using staples or tags. The arrangement of data in topics, using abbreviations, is the next step. The abbreviated topics are written down either on the margins of the relevant data or on slips of paper which may be attached to the relevant pages. The process of classifying or labelling various kinds of data help in the preparation of a data index. Sometimes there are large data. In such situations, computers are helpful in developing systematic and comprehensive classification schemes using code numbers for different categories and sub-categories. The computerized classification system permits the use of organised data by several groups of people over a long period of time. It permits easy cross-classification and cross- comparison of descriptive narrations for complex analysis.

1.4.2 Analysis of Qualitative Data

Analysis of qualitative data means studying the organised material available in the form of detailed descriptions, direct quotations or case-documentation in order to discover inherent facts. These data are studied from as many angles as possible either to explore the new facts or to reinterpret already known or existing facts.

The following methods are generally used in the analysis of qualitative data:

i. **Content Analysis**

ii. **Inductive Analysis**

iii. **Logical Analysis**

A. Content Analysis

Content analysis is concerned with the classification, organisation and comparison of the content of the document or communication. The terms, content analysis and coding are sometimes used interchangeably as both the processes involve objective, systematic, and qualitative description of any symbolic behaviour. Since content analysis involves the classification, evaluation and comparison of the content of communication or document, it is sometimes referred to as ‘documentary activity’ or ‘information analysis’. The communication may be in the form of responses to an open-ended questionnaire, conversation as a result of an interview, or description of an observed activity. It may also be in the form of official records (census, birth, accident, crime, school, institutional and personal records), judicial decisions, laws, budget and financial records, cumulative records, courses of study, content of text books, reference works, news papers, periodicals or journals, prospectus of various educational institutions or universities, direct quotations, and notes of an interview.

There are three approaches that a researcher may adopt in content analysis. These include: (i) characteristics of content, (ii) procedures or causes of content, and (iii) audience or effects of content. In the first approach, the researcher is interested primarily in the characteristics of the content itself. He/she may focus either on the ‘substantive nature’ of the content or upon the ‘form’ of the content. In the second approach, the

researcher attempts to draw valid inferences about the nature of the procedures of the content or the causes of the symbolic material from the characteristics of the material itself. In the third approach to content analysis, the researcher interprets the content so as to reveal something about the nature of its 'audience' or its 'effects'. He/she takes the content material as a basis for drawing inference about the characteristics of the 'audience' for whom the material (content) is designed or about the effects of communication, which it brings about.

The steps involved in the process of content analysis includes (i) defining the unit of analysis, (ii) specifying variables and categories, (iii) frequency, direction and intensity of units, (iv) contingency analysis, (v) sampling of units, and (vi) constructing the content analysis outline. Defining the unit of analysis indicates whether the unit (material) is confined to single words, phrases, complete sentences, paragraphs, or to even larger amounts of materials. Once the unit is defined, the researcher conducts its analysis so as to create reproducible or objective data for scientific treatment and generalisation beyond the specific set of symbolic material analysed. For converting symbolic material into objective data, it is necessary to specify the "variables" explicitly in terms of which descriptions are to be made. Once the unit is defined and the variables along with their categories specified, the researcher will classify units in the material to be analysed according to : (i) the number of units (frequency), (ii) favourableness/ unfavourableness of the content (direction), and (iii) the emotional impact of the units (intensity). The contingency analysis aims at considering the content within which the unit is found. The researcher considers the favourableness or unfavourableness of a single unit in the light of the reminder of the communication so that its real meaning is not lost.

Steps in the Analysis of Qualitative Data.

1. The first step is to understand your data. Must read the data carefully again and again for the quality.
2. Second step is to identify the purpose of evaluation. See that how the respondents have answered to the question.
3. Categorize the data into themes/patterns. Then organize into categories. This is the most important step in Qualitative Analysis. You can assign the codes which can be few letters or words or symbols.
4. In the next step you have to find out the patterns and connections within the categories or between the categories identified.
5. The last step is the interpreting the data. You have to think and design an outline to present the findings.
6. You can feed the data to the computer by entering the text to word processing program. These days software programmes like Ethnograph, MODIST etc. can be used to analyze the qualitative data. If the data is not big, you can also analyze manually. There are other software also available. You can choose according to your convenience.

B. Inductive Analysis

Inductive analysis means that patterns, themes, and categories of analysis emerge out of the data. In this analysis, researcher looks for natural variation in the data. The study of natural variation involves particular attention to variations in programme processes and how participants respond to and are affected by programmes. Two ways of representing the patterns emerge from the analysis of data. First, the researcher can use the categories developed and articulated in the programme studied to organize presentation of particular themes. Second, the researcher may also become aware of categories or patterns for which the people in the programme did not have labels or terms, and the analyst develops terms to describe these inductively generated categories.

C. Logical Analysis

Logical analysis is used for representing patterns as dimensions or categories using either participant-generated constructions or evaluator generated constructions. It is sometimes useful to cross-classify different dimensions to generate new insights about how the data can be organized and to look for patterns that may not have been recognised in the initial induction analysis. Logical analysis aims at creating potential categories by crossing one typology with another, and then moving back and forth between the logical construction and the actual data for creating a "new typology" using cross-classification matrices.

There are other ways of analysing qualitative data. We have not discussed all of them. The idea is to give you a feel of qualitative data analysis and show how it differs from quantitative data analysis.

Check Your Progress 6

Define qualitative data. Give some examples of these data.

Notes: (a) Space is given below for writing your answer.
(b) Compare your answer with the one given at the end of the Unit.

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1.5 LET US SUM UP

In this Unit, we discussed the nature of quantitative and qualitative data, the various methods of representing the quantified data graphically, and the methods used in the analysis of quantitative and qualitative data. The main points are as follows:

1. The data collected through the administration of various tools on the selected samples are of (i) quantitative and (ii) qualitative nature.
2. Quantitative data are expressed in nominal, ordinal, interval or ratio scales of measurement. These data are classified into two categories: (i) parametric and (ii) non-parametric. The parametric data are obtained by applying interval or ratio scales of measurement, whereas non-parametric data are either enumerated or ranked. In the enumerated data we make use of nominal scale and in the ranked one we apply ordinal scale.
3. The quantified data is tabulated in 'frequency distribution' and can be represented graphically with the help of a histogram, a frequency polygon, and/or an ogive.
4. Measures of (i) central tendency, (ii) variability, (iii) relative positions, and (iv) relationship are the four types of descriptive statistical measures.
5. Mean, median and mode are the three measures of central tendency.
6. Mean is the arithmetic average of a distribution. It is obtained by dividing the sum of all values of observation by the total number of values. The formula for finding the mean for ungrouped data is:

$$M = \frac{\sum X}{N}$$

When the number of observations is large, the data is grouped in a frequency distribution. The formula for computing the mean here is:

$$M = AM + \frac{\sum fx}{N} \times i$$

7. Median is a point in an array, above and below which one half of the values or measures fall. If the values are ungrouped and their number is small, the values are arranged in order of magnitude and the middle value is determined by counting up half the value of N.

When the number of values is odd, the mid-value is the median. When the number of values is even, the median is the mid-point between the two middle values.

In the case of grouped data, the median is calculated by the formula:

$$Mdn = l + \frac{\frac{N}{2} - F}{x} xi$$

8. Mode is the most frequently occurring value in a distribution. If only one value occurs a maximum number of times, the distribution is said to have one mode (uni-modal). A two mode distribution is bi-modal, and more than a two mode distribution is called multimodal.

In a simple ungrouped series of measures or values, the crude mode is that single measure or value which occurs most frequently.

For a group distribution, the mode is calculated by the formula:

$$\text{Mode} = l + \frac{fm - fi}{2\,fm - f_1 - f_2}\,xi$$

9. The range, variance and standard deviation are the most commonly used measures of variability.
10. The range is the difference between the two extreme values or measures in a distribution.
11. The average of the squared deviations of the measures or values from their mean is known as variance. Standard deviation is the positive square root of variance. Variance and standard deviation for the ungrouped data are found by the formulae:

$$\text{V}ariance = \sigma^2 = + \frac{N\sum X^2 - \left(\sum X\right)^2}{N^2}$$

$$\text{Standard Deviation } \sigma = \sqrt{\frac{N\sum X^2 - \left(\sum X\right)^2}{N^2}}$$

When the data are grouped in a frequency distribution, the variance and standard deviation are computed by the formulae:

$$\text{V}ariance = \sigma^2 = \frac{i^2}{N^2} + \left[N\sum fx'^2 - \left(\sum fx'\right)^2\right]$$

$$\text{S}tandard\ \text{Deviation} = \sigma = \frac{i}{N}\sqrt{\left[N\sum fx'^2 - \left(\sum fx'\right)^2\right]}$$

12. The normal probability distribution is represented by a curve which has the following characteristics:
    i) The curve is symmetrical around its vertical axis called ordinate.
    ii) The mean, mode and median of the distribution have the same values.
    iii) The height of the vertical line called ordinate is maximum at the mean and in the unit normal curve, it is equal to 0.3989.
    iv) The curve is asymptotic.
    v) The points of inflection of the curve occur at points 1 ± , standard deviation ($\pm 1\sigma$) above and below the mean.
    vi) About 68.26 percent of the total area of the curve falls between limits Mean $\pm 1\sigma$, 95.44 percent of the total area falls between $M \pm 2\sigma$ and 99.73 percent of the total area falls between $M \pm 3\sigma$.

13. Sigma scores, standard scores, percentiles and percentile ranks are the measures of relative positions.
14. A sigma score makes it possible to obtain a realistic comparison of scores and provides a basis for equal weighting of the scores as the scores on different tests are expressed on a scale with a mean of zero and standard deviation 1.
15. When the sigma scores are converted into a new distribution with mean and standard deviation so selected as to make all scores positive, the scores are called standard scores.

The formula for the conversion of a raw score to a standard score is:

$$X = \frac{\sigma'}{\sigma}(X - M) + M'$$

When the mean (M) and standard deviation (σ') are taken to be 50 and 10 respectively, the standard score is called a T-score. It is expressed by the formula:

$$T = \frac{10}{\sigma}(X - M) + 50$$

16. Percentiles are the points which divide the entire scale of measurement into 100 equal parts.

    The formula for computing percentiles is:

$$P_p = l + \frac{(P - F)}{f} \times i$$

17. Percentile rank is the point in the distribution below which a given percentage of scores fall. If R is the rank and N is the total number of cases.

$$\text{Percentile Rank} = 100 - \frac{100R - 50}{N}$$

18. Product Moment correlation and rank-difference correlation are the commonly used measures of relationship between any two variables.
19. When the size of sample is small and the variables are measured in interval scales of measurement, the product-moment correlation is computed by the formulae:

$$r_{xy} = \frac{N\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{[N\Sigma x^2 - (\Sigma x)^2[(N\Sigma y^2 - (\Sigma y)^2]}}$$

When the size of the sample is large, the product-moment correlation is found by the formulae:

$$r_{xy} = \frac{N\sum fxy - (\sum fx)(fy)}{\sqrt{[N\sum fx^2 - (\sum fx)^2[(N\sum fy^2-(\sum fy)^2]}}$$

20. When the data are available in ordinal (rank) form of measurement and the size of the sample is small, the formula for computing the rank-difference correlation is:

$$\rho = 1 - \frac{6\sum D^2}{N(N^2 - 1)}$$

21. In some situations if it is difficult to measure or analyse a phenomenon into various components or variables in quantified terms, the researcher takes into consideration the phenomenon as a whole, in detail and depth. In other words, the researcher uses qualitative techniques of analysis. Qualitative data consist of detailed descriptions of situations, events, people, interactions, and observed behaviours. These data are also available in the form of direct quotations from people about their experiences, attitudes, beliefs, and thoughts. The excerpts or entire passages from documents, correspondence, records and case studies are also examples of qualitative data.
22. Content analysis, inductive analysis and logical analysis are some methods of qualitative analysis.
23. Content analysis pertains to the classification, quantification and consideration. It is also called documentary or information analysis.
24. Inductive analysis leads to patterns, themes and categories emerging out of the data. In this type of analysis, the researcher looks for natural variation in the data.
25. Logical analysis is used for representing patterns as dimensions or categories, either using participant-generated constructions or evaluator-generated constructions.

To assess your learning yourself, see whether you are now able to:

- name various types of data
- define quantitative data
- describe quantitative data
- describe various types of quantitative data with examples
- tabulate a given quantitative data into frequency distribution
- illustrate the methods of expressing the class intervals with the help of examples
- compute the cumulative frequencies and cumulative percents for a given frequency distribution
- name the four methods of representing a data graphically
- construct a histogram, a frequency polygon and an ogive for a given distribution
- name the four descriptive statistical measures
- name and define the three measures of central tendency or averages (mean, median and mode)
- compute mean, median and mode from a given (i) ungrouped data (ii) grouped data

- name and define the three measures of variability (range, variance and standard deviation)
- compute range for ungrouped data
- compute variance and standard deviation for a given (i) ungrouped and (ii) grouped data
- describe the nature and characteristics of Normal Distribution
- use the Normal Table
- name and define the various measures of comparing individuals on the basis of different types of scores (Sigma Scores, Standard Scores, Percentiles and Percentile ranks)
- convert a raw score into a sigma score corresponding to the mean and standard deviation of a distribution
- convert a given raw score into a standard score (Z-score or T-score) corresponding to the mean and standard deviation of a distribution
- define a percentile
- compute certain percentiles for a given distribution of scores
- define percentile rank
- compute the percentile rank of an individual corresponding to his/her rank in the group to which he/she belongs
- name the various measures of relationships
- compute the product moment correlation for a given (i) ungrouped and (ii) grouped data
- compute rank order correlation for a given ungrouped data
- define qualitative data with examples
- name and describe some methods used in the analysis of qualitative data

1.6 GLOSSARY

1. Quantitative Data:	Data which are expressed in nominal, ordinal, interval or ratio scales of measurement.
2. Qualitative Data:	Data which are available in the form of detailed descriptions of situations, events, people, interactions, and observed behaviour, direct quotations from people about their experiences, attitudes, beliefs, and thoughts, and excerpts from documents, correspondence, records, and case histories.
3. Parametric Data:	These are data which are got by applying interval or ratio scales of measurement.
4. Nonparametric Data:	These are data which are got by applying nominal or ordinal scales of measurement. These types of data are either counted or ranked.
5. Central Tendency:	A measure of central tendency provides a single most typical value as representative of a group of

	values; the 'trend' of a group of measures as indicated by some type of averages, usually the mean, median or mode.
6. Mean:	A kind of average obtained by dividing the sum of a set of measures by their number.
7. Median :	The middle value in a distribution or set of ranked values; the point that divides the group into two equal parts.
8. Mode:	The value that occurs most frequently in a distribution.
9. Variability:	The spread or dispersion of measures or values.
10. Range:	For some specified groups, the difference between the highest and the lowest obtained measure or value on a tool. It is a rough measure of variability.
11. Variance:	A measure of variability of a distribution. It is the average of the squared deviations of the measures or values from the mean.
12. Standard Deviation:	The positive square root of variance.
13. Standard Score:	A general term referring to any of the variety of 'transformed' scores, in terms of which raw scores may be expressed for reasons of convenience, comparability, ease of interpretation, etc. Sigma Scores, T-Scores etc. are the examples of standard score.
14. Normal Distribution:	A distribution of measures that in graphic form has a distinctive bell-shaped appearance. It is symmetrical and asymptotic. The mean, mode and median for this type of distribution have equal values.
15. Percentile Rank:	The expression of an obtained test score in terms of its position within a group of 100 scores.
16. Co-efficient of correlation:	A measure of the degree of relationship between two sets of measures for the same group of individuals. Its value ranges from 00, denoting a complete absence of relationship, to+1.00 and -1.00, indicating perfect positive and negative correspondence respectively.

1.7 CHECK YOUR PROGRESS: THE KEY

1. Quantitative data is the description of an empirical event or phenomenon in a numerical system presented with the help of different scales of measurement such as nominal, ordinal, interval and ratio. The two major types of quantitative data are: parametric (obtained through interval or ratio scales) and non-parametric (counted by a nominal scale

or ranked by a ordinal scale).

2&3 Class Interval	Mid Point	f		fx'	fx'^2
195-199	197	1	5	5	25
190-194	192	2	4	8	32
185-189	187	4	3	12	36
180-184	182	5	2	10	20
175-179	177	8	1	8	8
170-174	172	10	0	0	0
165-169	167	6	x' 1	-6	6
160-164	162	4	-2	-8	16
155-159	157	4	-3	-12	36
150-154	152	2	-4	-8	32
145-149	147	3	-5	-15	75
140-144	142	1	-6	-6	36
		N=50		$\sum fx' = -12$	$\sum fx'^2 = 322$

(i) Mean $= \text{A.M.} + \dfrac{\sum fx'}{N} \times i$

$= 172 + \dfrac{(-12)}{50} \times 5$

$= 170.80$

(ii) Variance $= \sigma^2 = \dfrac{i^2}{N}\left[N \sum fx'^2 - (\sum fx')^2\right]$

$= \dfrac{(5)^2}{(50)^2}\ [(50 \times 322 - (-12)^2]$

$= 159.52$

(iii) Standard Deviation $= \dfrac{i^2}{N}\sqrt{\left[N \sum fx'^2 - (\sum fx')^2\right]}$

$= \dfrac{5}{50}\sqrt{[(50 \times 322 - (-12)^2]}$

$= 12.63$

4 Normal probability curve is symmetrical around its vertical axis, i.e. ordinate.

The values of mean median and mode coincide and have the same value.

The height ordinate is maximum at the mean,

The curve is asymptotic.

The points of inflection of the curve occur at points 1 ± , standard deviation above and below the mean.

About 68.26 percent of the total area falls between the limits $M + 1\sigma$ and $M - 1\sigma$; 95.44 per cent of the total area of the curve falls between limits $M + 2\sigma$ and $M - 2\sigma$; and 99.73 percent of the total area of the curve falls between $M + 3\sigma$ and $M - 3\sigma$.

5.

X	Y	x	y	x2	y2	xy
45	56	-20	-9	400	81	180
55	50	-10	-15	100	225	150
56	48	-9	-17	81	289	153
58	60	-7	-5	49	25	35
60	62	-5	-3	25	9	15
65(AM)	64	0	-1	0	1	0
68	65(AM)	3	0	9	0	0
70	70	5	5	25	25	25
75	74	10	9	100	81	90
80	82	15	17	225	289	225
85	90	20	25	400	625	500
				$\sum x = 2,$	$\sum y = 6,$ $\sum x^2 = 14/4,$	$\sum y^2 = 1650,$ $\sum xy = 1403,$

$$r_{xy} = \frac{N\sum xy - (\sum x)(\sum y)}{\sqrt{[N\sum x^2 - (\sum x)^2[(N\sum y^2-(\sum y)^2]}}$$

$$= \frac{11 \times 1403 - 2 \times 6}{\sqrt{[11 \times 1414 - (2)^2[(11\times 1650 -(6)^2]}}$$

$$= 0.92$$

6 Qualitative data describes a phenomenon which cannot be measured or quantified. The phenomenon is looked at in its totality. Detailed descriptions of situations, events, people, interactions, and observed behaviours constitute qualitative data.

UNIT 4 DATA PROCESSING AND ANALYSIS

Structure

4.1 Introduction

4.2 Data Measurement and Its Type

4.3 Tabulation and Interpretation of Data

4.4 Let Us Sum Up

4.5 Keywords

4.6 References and Selected Readings

4.7 Check Your Progress – Possible Answers

4.1 INTRODUCTION

The purpose of data analysis is to identify whether research assumptions were correct or not, and to highlight possible new views on the problem under study. The ultimate purpose of analysis is to answer the research questions outlined in the objectives with the collected data. However, before we look at how variables may be affecting one another, we need to summarize the information obtained on each variable in simple, tabular form, or, in a figure.

Some of the variables may produce numerical (continuous) data, while other variables produce categorical data. In analyzing our data, it is important, first, to determine the type of data that we are dealing with. This is crucial because the type of data used largely determines the type of statistical techniques that should be used to analyze the data. Once the data is processed, tables and graphs are prepared, and the report writing work may be initiated.

After studying this unit, you should be able to

- define data and describe various types and nature of data.
- describe techniques of data processing, tabulation, and presentation.
- describe and interpret data from tables that have been generated.

4.2 DATA MEASUREMENT AND ITS TYPES

4.2.1 Data Measurement

Measurement is the process of observing and recording the observations that are collected as part of a research effort and the process of assigning numbers to objects or observations. But do we measure abstract concepts like happiness, quality of life, personality, opinion, etc? You have to understand that the qualities of good measurement are: (i) Precise; (ii) Unambiguous; (iii) Free from errors; (iv) Valid; (v) Reliable, and; (vi) Practical.

The tools, which are developed for measurement/ collection of data, should be

- valid to measure the characteristic which it is intended to measure
- reliable to the extent to which an experiment/ test/ or any measuring procedure yields the same result on repeated trials

- sensitive enough to detect differences in a characteristics
- specific enough to represent only the characteristic of interest
- appropriate to the objectives of the study
- provides adequate distribution of response in the study population
- meets the objective of the study

These are key points to keep in mind while developing effective tools for data collection. However, the following factors may affect the validity of the data, so remedial steps should be taken accordingly:

Respondents: reluctance, modesty, having little knowledge about details of the research problem to answer, may not admit ignorance, guessing about response, feel boredom due to long questioning, fatigue, and anxiety, etc.

Situation: lack of support to field investigators, lack of assurance on anonymity and confidentiality, etc.

Investigator/ Interviewer: may reword or reorder questions, style, and look, carelessness in filling the reply, incorrect coding, or calculation of scores.

Instrument: complex words, ambiguous meaning, poor printing, inadequate space for answer, response choice omission.

4.2.2 Types of Data

Data can be classified as either

i) Primary or secondary data

ii) Time series, cross section and panel data

iii) Categorical and numerical data

i) Primary or Secondary Data

Primary Data

Primary data means original data that have been collected specially for the purpose in mind. Research where one gathers this kind of data is referred to as field research. Tools used for gathering primary data, for example, a questionnaire, an interview schedule etc.

Secondary Data

Secondary data are data that have been collected for another purpose and where we will use a statistical method with the primary data. This means that after performing statistical operations on primary data the results become known as secondary data. Research where one gathers this kind of data is referred to as desk research. The source for gathering secondary data, for example data from a book.

ii) Time series, Cross-Section and Panel Data

Cross-sectional data refers to data collected by observing many subjects (such as individuals, firms or countries/regions) at the same point of time, or, without regard to differences in time. Analysis of cross-sectional data usually consists of

comparing the differences among the subjects. For example, we want to measure current obesity levels in a population. The cross-sectional data provides us with a snapshot of that population, at that one point in time. Note that we do not know, based on one cross-sectional sample, if obesity is increasing or decreasing; we can only describe the current proportion.

Time series data is also known as longitudinal data, which follows one subject's changes over the course of time. For example, the average production of wheat from 1990 to 2009.

Panel data combines both cross sectional and time series data and looks at multiple subjects and how they change over the course of time. Panel analysis uses panel data to examine changes in variables over time, and differences in variables between subjects.

iii) Categorical and Numerical Data

Categorical or Nominal: Data that can be divided into categories or groups such as male and female, and can take only discrete and not decimal values. They are called categorical or nominal data. There are two types of categorical data: they are nominal or ordinal. In nominal data, the variables are divided into a number of named categories. These categories, however, cannot be ordered one above another (as they are not greater or lesser than each other).

Continuous or Numerical: Data that can take any value, including decimal values, are called continuous data. Data that can be measured on a scale is said to be scalar. We speak of numerical data if they are expressed in numbers. There are two types of numerical data: they are discrete or continuous. Discrete data are a distinct series of numbers.

In this section, you studied about the data measurement and its types. Now answer the following questions.

Check Your Progress 1

Note: a) Write your answer in about 50 words.

b) Check your answer with possible answers given at the end of the unit

1) What is panel data?

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2) What do you understand by primary data?

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4.3 TABULATION AND INTERPRETATION OF DATA

The following operations need to be done to bring data into a presentable form:

- Data coding, editing and feeding
- Data tabulation
- Figures and graphs

We will now discuss these operations, one after the other, for both categorical and numerical data.

4.3.1 Data Coding, Editing and Feeding

Once data is collected the researcher thinks about the processing and analysis of the data. This is a crucial stage of research work. Here, the researcher has to consult the guide and other academicians who have done research in the related field. Therefore, it is advised that data processing must be planned in advance, and, necessarily, during the time of questionnaire formulation; and, during the time of data collection. Nowadays, with the advent of the computer, researchers might think it is the job of computer assistant, but it is not so. Before either giving it to the computer assistant, or doing it manually, a lot of activities have to be completed: (i) editing of the data which you received from the field; (ii) coding of the data; (iii) preparation of the master chart if the data is to be computed manually; and (iv) presentation of data.

i) Editing of the data

The editing of data is the first step of data processing and analysis. After collecting data, either through the questionnaire or schedule, you have to edit it. You have to carefully check for missing and wrongly entered data. Therefore, in large scale surveys and research, the company undertaking the research project appoints supervisors, or, editors, for proper checking of data. An example of editing of data is given below.

Example – Suppose that you conduct a survey of the occupation status of the people in a village, for which you have to collect data from heads of households. The questionnaire for the purpose is as follows

Name of the Respondent:	-	Rama Singh
Age:	-	56
Sex:	-	Male
Cast:	-	General
Occupation:	-	Private Service
Income of the Respondent:	-	10,000
Wife's name:	-	Rita Singh
Age:	-	51
Occupation:	-	House wife
Monthly Income:	-	800/-

In this questionnaire, the wife of Rama Singh is a housewife, but she earns Rs. 800. She may be a housewife, but by knitting she may be earning Rs. 800 per month. Therefore, her occupation may be categorized as self-employment. Like this, you have to cross check the data and try to get accurate information which will minimize error and strengthen your research findings. Therefore editing of the data is very essential before sending it for processing.

ii) Coding of the data

The questionnaire must be properly coded. Before either, sending it for feeding into the computer, or entering it in the master chart, the coding of data is necessary. The coding of data will make data entry easy. Coding of data means assigning numerical symbol to each response of the question. The purpose of giving numerical symbols is to translate raw data into numerical data, which may be counted and tabulated. An example of coding of the marital status and education is given below.

Marital Status

Married: - 01

Separated: - 02

Widow: - 03

Divorced: - 05

Never Married: - 06

Any Other: - 07

The coding of data and checking that codes are properly inserted in the questionnaire must be done during the editing of the questionnaire. It is always better to prepare a code book for your questionnaire. The coding can also be done during the time of data collection, if the code book is available with you.

iii) Entering data in the Master Chart

If you are doing tabulation manually, it is always wise to enter the data into a master chart. The master chart is a large sheet which will enable you to enter all the codes of different variables into it. It will help you to generate tables easily. An example of master chart is given below.

Master Chart

Sl.No	Name of the respondent	Variables					
		Sex	Age	Marital	Status	Education	Occupation
1							
2							
3							
4							
5							
	Total						

In the master chart you can enter the data of 14 sample respondents. Likewise you can expand the number of respondents in the column and variables in the rows. It is always better to enter code (numerical number) in the master chart.

iv) Entering Data into the Computer

Computers are widely used for the analysis of data. It makes the calculation much faster. The excel sheet, and the SPSS package can be used in social science research. The following steps are used in the analysis of data by using the SPSS package.

i) Entering of data in the SPSS statistical package

ii) Selection of procedure from the menu

iii) Selection of variables for analysis

iv) Examination of the output.

v) Presentation of Statistical Data

Statistical data are collected to serve a purpose. Therefore, data may be presented in such a way that it may be easily grasped and the conclusion may be drawn promptly. Generally, the following three methods are used for the presentation of statistical information.

a) **Textual Presentation**: In this method, statistical information is presented in text form. Generally, this type of presentation is made in a descriptive way. It requires careful reading of the text in order to grasp the meaning and significance of the facts and figures given, therein. But, for most people, it is not a suitable and effective method of presentation of statistical information, because it is not easy for the reader to single out the individual information and figures. The advantage of this method is that an ordinary person can prepare and present the text and a layman can read it and grasp it.

b) **Tabular Presentation**: In this method, statistical information is presented in the form of a table. Facts and figures are gathered, and then, incorporated in tables. Generally, this type of presentation is made in a tabular form with rows and columns. Tables summarize statistical data in a logical and orderly manner. The main advantage of this method is that tables are brief, concise, and contain only relevant figures. Tables also facilitate the comparison of figures. The only disadvantage of this method is that the presentation of tables and the interpretation of data require some skills and techniques.

c) **Graphical Presentation**: In this case, statistical information is presented in the form of graphs and charts. Facts and figures are gathered first, and then, they are depicted in graphs and charts for presentation. Generally, this type of presentation is made through figures, diagrams, charts, or graphs. The main advantage of this method is that the facts and figures become more attractive and appealing to eye. A disadvantage of this method is that facts cannot be shown in detail and accurately.

4.3.2 Data Tabulation

Tabulation is an orderly and systematic arrangement of numerical data presented in rows and columns for the purpose of information, comparison, and interpretation. So, a statistical table is a systematic arrangement of statistical

data into rows and columns. It summarizes the data in a logical and orderly manner for the purpose of presentation, comparison, and interpretation. Tabulation is, thus, a scientific process and means of recording, statistical data in a systematic and orderly manner.

A statistical table has the following five parts.

i) **Title:** each table must have a title which convey the contents of the table. It should be clear, concise and self-explanatory. It should be written on the top of the table.

ii) **Stub:** this is a column used for mentioning the items and their heading. It is the left most column of the table. A stub is generally marked with rows and in each row, an item is mentioned. The stub should be clear and self-explanatory.

iii) **Caption:** this is the heading for columns, other than the stub. It is the upper part of the Table. Caption should be properly columned and worded. Sometimes, below the caption, the units of measurement and column-numbers, is called a box head.

iv) **Body:** the main part of the table. It contains the data which are exhibited in the table. The figures inserted therein should be distinct.

v) **Source & Footnote:** the last part of a table. If the researcher is procuring data from a secondary source, then the source of the data needs to be mentioned. For example, if you are citing data from the Census of India then the year and the department and the state need to be mentioned. After mentioning the sources, the researcher has to provide a footnote, e.g., in the same cell, if you are giving the figure, and in parenthesis, the percentage to the total, and then, it must be mentioned in the footnote. An example of a table is given below:

Title

		Caption
STUB	**BODY**	**BOX HEAD**

Source:

Footnote:

4.3.3 Types of Tabulation

Tabulation is done based on the data. The following types of tables are generally constructed.

i) Construction of frequency distribution table

ii) Construction of cross- tabulation

iii) Construction of figures and graph

i) Construction of Frequency Distribution Table

A frequency distribution table can be of two types:

- Simple frequency distribution
- Grouped frequency distribution

In constructing a simple frequency distribution, the observations are not divided into groups or classes. Only individual values are shown, whereas, in the grouped frequency distribution the observations are divided into groups or classes.

Here is a simple frequency distribution table.

Table 4.1: Mark obtained by class fifth students in mathematics

Marks	Tally of marks	Frequency (No. of students)
20	𝍸	5
21	𝍸 II	12
25	𝍸 III	13
30	𝍸	10
Total		**40**

Table 4.2: Grouped frequency distribution

Marks obtained	Tally of marks	Frequency
1-10	𝍸 I	11
11-20	𝍸 𝍸 IIII	20
21-30	𝍸 𝍸 IIII	21
31-40	𝍸 III	13
41-50	𝍸 IIII	9

While preparing the group frequency distributions, the following points have to be taken into consideration.

- The groups must not overlap, otherwise there will be confusion about which group a measurement belongs to.
- There must be continuity from one group to the next, which means that there must be no gaps. Otherwise, some measurements may not fit in a group.
- The groups must range from the lowest measurement to the highest measurement so that all of the measurements have a group to which they can be assigned.
- The groups should normally be of an equal width, so that the counts in different groups can easily be compared.

i) Construction of Cross Tabulation

So far, we have made tables containing frequency distributions for one variable at a time, in order to partially describe our data. Depending on the objectives of our study, and the study type, we may have to examine the relationship between

several of our variables at the same time. For this purpose it is appropriate to construct cross tabulation of data. Depending on the objectives and the type of study, different kinds of cross tabulations may be required. The examples of cross tabulation are given below. Here, three different types of cross tabulation of data have been given.

Example 1: A study was carried out on the degree of job satisfaction among doctors and nurses in rural and urban areas. To describe the sample, a cross tabulation was constructed which included the sex and the residence (rural, or urban) of the doctors and nurses interviewed. This was useful, because in the analysis, the opinions of male and female staff had to be compared separately for rural and urban areas.

Table 4.3: Type of teachers by residence

Residence	Type of teachers		
	Principles	**Teachers**	**Total**
Rural	10 (16%)	69 (38%)	79 (33%)
Urban	51 (84%)	113(62%)	164 (67%)
Total	61(100%)	182 (100%)	243(100%)

Interpretation: table 4.3 shows that a higher percentage of teachers than principals work in rural areas, but, that, overall, a greater proportion of teaching staff works in urban areas (67%).

Table 4.4: Sex of teachers by residence

Residence	Sex of teachers		
	Male	**Female**	**Total**
Rural	54 (43%)	25 (21%)	79 (33%)
Urban	71 (57%)	93 (79%)	164 (67%)
Total	125(100%)	118 (100%)	243(100%)

Interpretation: it can be concluded from Table 4.4 that there are more males serving in rural areas than females. These males in rural areas are apparently teachers.

To obtain an overview of the distribution of principals and teachers by gender in rural and urban areas, we can construct the following two-by-four cross-table.

Table 4.5: Residence and sex of principals and teachers

Teaching staff		Residence		
		Rural	**Urban**	**Total**
Principals	Males	8 (10%)	35 (21%)	43 (18%)
	Females	2 (3%)	16 (10%)	18 (7%)
Teachers	Males	46 (58%)	36 (22%)	82 (34%)
	Females	23 (29%)	77 (47%)	100 (41%)
	Total	79(100%)	164 (100%)	243(100%)

Interpretation: this table shows, in a glance, that male nurses dominate the rural health services. It also indicates that males dominate in the medical profession, (18% M > 7% F doctors) but, that, overall, there are more female than male nurses, and that the females are mainly clustered in towns.

The data in the tables is usually listed in absolute figures, as well as in relative frequencies (percentages or proportions). As already seen in Unit 3, for numerical data (such as age) the mean, median, and/or mode, with standard deviation may be calculated as well, to describe the sample.

General hints while constructing tables

- Make sure that all the categories of the variables presented in the tables have been specified, and that they are mutually exclusive (i.e., no overlaps and no gaps) and are exhaustive.
- When making cross-tabulations, check that the column and row counts correspond to the frequency counts for each variable.
- Also, check that the grand total in the table corresponds to the number of subjects in the sample. If not, an explanation is required. This could be presented as a footnote. (missing data, for example.)
- Think of a clear title for each table. Also, be sure that the headings of rows and columns leave no room for misinterpretation.
- Number your tables and graphs and keep them together with the objectives to which they are related. [Numbering of tables and graphs in a chapter (e.g., 4) may be like table 4.1, table 4.2 and Figure 4.1 Figure 4.2, etc.] This will assist in organizing your report and ensure that work is not duplicated.

i) Construction of figures and graph

If your report contains many descriptive tables, it may be more readable, if you present the most important ones in figures. The most frequently used figures for presenting data include

• Bar charts	for categorical data
• Pie charts	
• Histograms	for continuous data
• Line graphs	
• Scatter diagrams	
• Maps	

We will now look at example of the abovementioned figures that can be used for presenting data.

1) Bar Chart

The data from Example 2 can be presented in a bar chart, using either absolute frequencies or relative frequencies/percentages and an example is given in Figure 4.1 below.

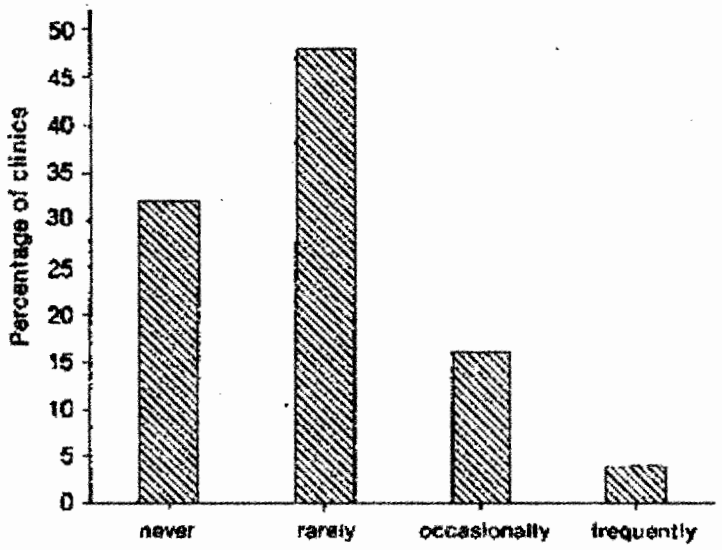


Figure 4.1: Relative frequency of shortage of anti-malaria drugs in rural health institutions (n=148)

Note that the sample size must be indicated if you present the data in percentages.

2) Pie Charts

A pie chart can be used for the same set of data, providing the reader with a quick overview of the data presented in a different form. A pie chart illustrates the relative frequency of a number of items. All the segments of the pie chart should add up to 100%.

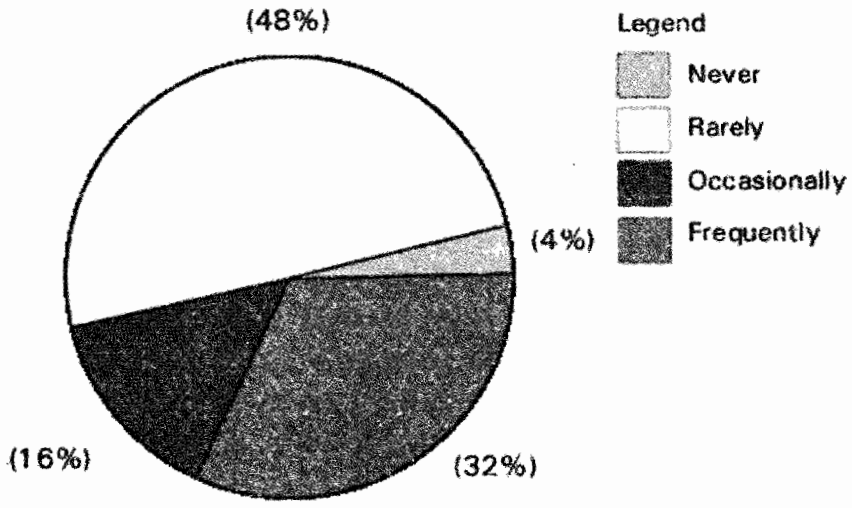


Figure 4.2: Relative frequency of shortage of anti-malaria drugs in rural health institutions (n=148)

3) Histograms

Numerical data are often presented in histograms, which are very similar to the bar charts which are used for categorical data. An important difference, however, is that in a histogram the bars are connected (as long as there is no gap between the data), whereas in a bar chart the bars are not connected, as the different categories are distinct entities. An example of histogram is given in Figure 4.3.

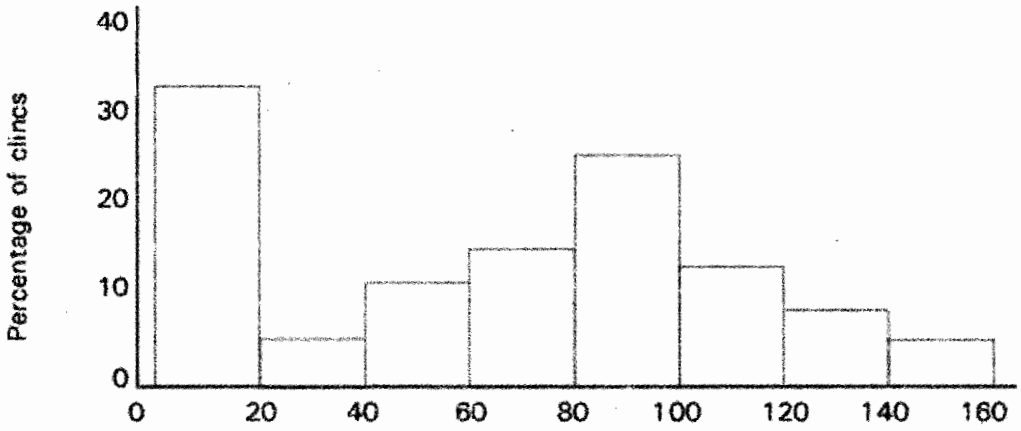


Figure 4.3: Percentage of clinics treating different numbers of malaria patients in one month (n=80).

No. of patients per month

4) Line Graphs

A line graph is particularly useful for numerical data if you wish to show a trend over time. An example of a line graph is given in Figure 4.4.

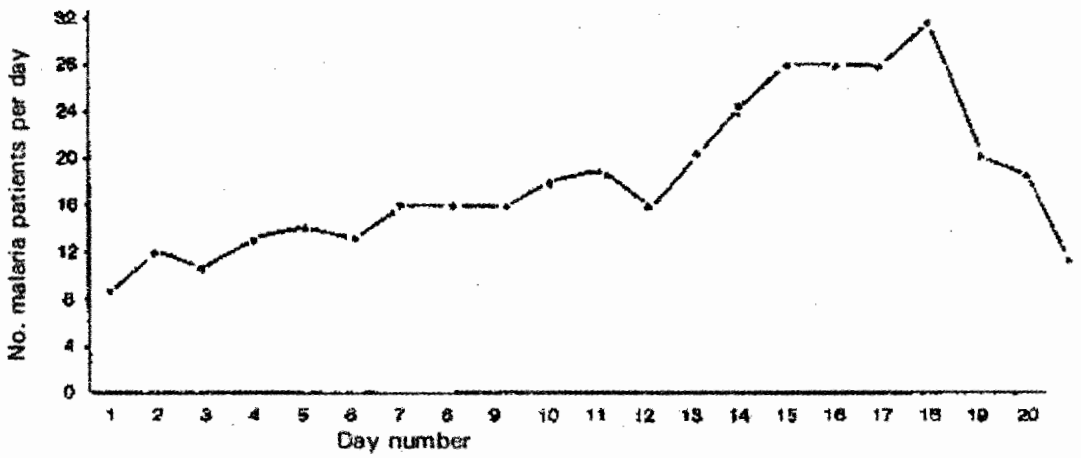


Figure 4.4: Daily number of malaria patients at the health centres in District X

It is easy to show two or more distributions in one graph, as long as the difference between the lines is easy to distinguish. Thus, it is possible to compare frequency distributions of different groups, i.e., the age distribution between males and females, or cases and controls.

5) Scatter diagrams

Scatter diagrams are useful for showing information on two variables which are possibly related. The example of a scatter diagram, given below, is used where we are dealing with the concepts of association and correlation.

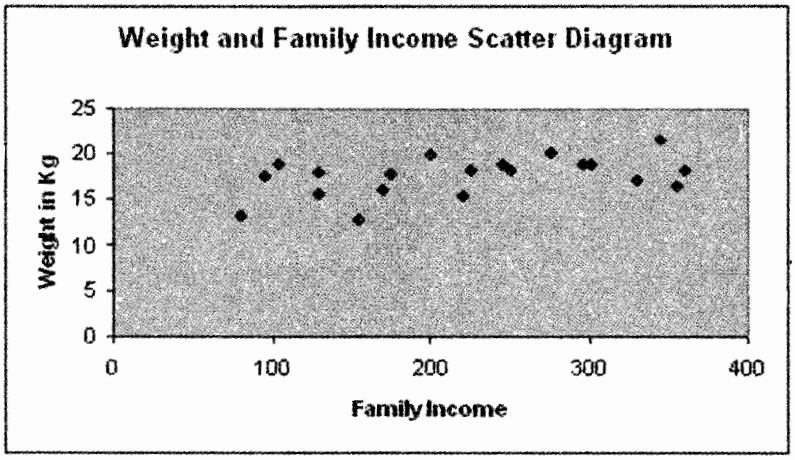


Figure 4.5: Weight of five-year-olds according to annual family income

Note: It is important that all figures presented in your research report have numbers, clear titles, and are clearly labelled (or keyed).

6) Maps

In addition to the figures above, the use of maps may be considered to present information. For instance, the area, where a study was carried out, can be shown in a map. If the study explored the epidemiology of cholera, a map could be produced showing the geographical distribution of cholera cases, together with the distribution of protected water sources, thus illustrating that there is an association. If the study related to vaccination coverage, a map could be developed to indicate the clinic sites and the vaccination coverage among under-fives in each village, perhaps showing that home-clinic distance is an important factor associated with vaccination status.

In this section, we discussed about the tabulation and interpretation of data. Now answer the following questions.

Check Your Progress 2

Note: a) Write your answer in about 50 words.

b) Check your answer with possible answers given at the end of the unit

1) What is meant by coding of data?

.....

.....

.....

.....

2) What is a pie chart and where is it used?

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4.4 LET US SUM UP

Understanding data quality and its measurement are of utmost importance for any researcher. Poor quality data cannot be analysed properly, and, may also give results which may not be valid, or, sometimes give results which could have adverse consequences for society if those results are used as a base for policy making. The presentation of data needs to be made in simple, as well as cross table format. It is always advisable to prepare a cross tabulation plan as per study requirements, and to present data in graphical way for clarity.

4.5 KEYWORDS

Data Measurement : measurement is the process of observing and recording the observations that are collected as part of a research effort.

Type of Data : broadly there are two types of data: (i) quantitative and; (ii) qualitative which can be further classified as categorical, nominal and continuous data.

Data Quality : the quality data can be characterized as: (i) precise, (ii) unambiguous, (iii) free from errors, (iv) valid, (v) reliable, and (vi) practical.

Data Processing : means the generation of frequency distribution and cross tabulation and calculation of other statistical measures.

Frequency Distribution : preparation of tables which distribute respondents according to a particular characteristic of sample, or research outcome.

Cross Tabulation : this is a process of generating tables giving the outcome of interest in columns, and various characteristics of respondents, or factors affecting outcomes in rows.

Data Interpretation : is drawing valid and meaningful conclusions from the tables generated with the help of collected data.

Report Preparation : is the process of documenting the whole process of research conducted to identify the problem, or to prove some relationships, or for proving the success of some programme related activities.

4.6 REFERENCES AND SELECTED READINGS

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4.7 CHECK YOUR PROGRESS - POSSIBLE ANSWERS

Check Your Progress 1

1) What is panel data.

Answer: Panel data combines both time series and cross sectional data and looks at multiple subjects and how they change over the course of time. Panel analysis uses panel data to examine changes in variables over time, and differences in variables between subjects.

2) What do you understand by primary data?

Answer: Primary data is original data that have been collected specially with a purpose in mind. Research where one gathers this kind of data is referred to as field research, for example: a questionnaire.

Check Your Progress 2

1) What is meant by coding of data?

Answer: A questionnaire must be properly coded. Before feeding it into a computer, or entering the data into the master chart, coding of data is necessary. The coding of data will make data entry easy. Coding of data means assigning a numerical symbol to each response of the question. The purpose of giving numerical symbols is to translate raw data into numerical data, which may be counted and tabulated.

2) What is a pie chart and where is it used?

Answer: A pie chart can be used for providing the reader with a quick overview of the data presented in a different form. A pie chart illustrates the relative frequency of a number of items. All the segments of the pie chart should add up to 100%.

Module - 1

Chapter - 1

A time series is a sequence of statistical data organized according to the time of occurrence or in chronological order. The numerical data collected at various points in time, forming a set of observations, is referred to as a time series. In time series analysis, current data within a series can be compared with past data from the same series. Additionally, the progression of two or more series over time can be compared. These comparisons can provide valuable insights for individual businesses. Time series analysis is crucial in fields such as economics, statistics, and commerce.

1 Time Series Data

A time series consists of observations made at specific time intervals and arranged in chronological order. For example, tracking agricultural production, sales, or National Income over a span of 3 to 5 years constitutes a time series. It is essentially a sequence of quantitative readings recorded at regular intervals, which could be hourly, daily, weekly, monthly, or annually. Examples of time series include hourly temperature readings, daily shop sales, weekly market sales, monthly production figures, yearly agricultural outputs, and population growth over ten years. Analyzing a time series involves comparing past data with current data to forecast future trends and evaluate past performance. The focus of time series analysis is on understanding chronological variations. Key requirements for a time series are:

- The time intervals between observations should be as consistent as possible.
- The dataset must be homogeneous.
- Data should be collected over an extended period.

Symbolically, if t represents time and y_t denotes the value at time t, then the paired values (t, y_t) constitute the time series data. Ex 1: Production of rice in Karnataka for the period from 2010-11 to 2016-17.

Table 1: Production of rice in Karnataka (in '000 metric tons)

Year	Production
2010-11	800
2011-12	950
2012-13	870
2013-14	920
2014-15	860
2016-17	720

1.1 Purpose

Time series analysis is crucial for understanding historical data and forecasting future trends, which aids managers and policymakers in making informed decisions. By quantifying key features and random variations in data, time series methods have become widely applicable across government, industry, and commerce, especially with advances in computing power. The Kyoto Protocol, an amendment to the United Nations Framework Convention on Climate Change, was signed in December 1997 and came into effect on February 16, 2005. The rationale for reducing greenhouse gas emissions involves a blend of scientific data, economic considerations, and time series analysis. The decisions made in the coming years will have significant implications for the planet's future.
In 2006, Singapore Airlines expanded its fleet by ordering twenty Boeing 787-9s and expressing intent to purchase twenty-nine Airbus planes, including twenty A350s and nine A380s (superjumbos). This expansion was guided by time series analysis of passenger trends and strategic corporate planning to maintain or enhance market share. Time

series methods are also employed in everyday operational decisions. For instance, UK gas suppliers must place orders for offshore gas one day in advance. The variation from the seasonal average is influenced by temperature and, to a lesser extent, wind speed. Time series analysis helps forecast demand by adjusting the seasonal average with one-day-ahead weather forecasts.
Additionally, time series models underpin many computer simulations. Examples include evaluating inventory control strategies using simulated demand series, comparing wave power device designs with simulated sea states, and simulating daily rainfall to assess the long-term environmental impacts of proposed water management policies.

1.2 Time series

In many fields, including science, engineering, and commerce, variables are measured sequentially over time. For instance, reserve banks track daily interest and exchange rates, governments report annual GDP figures, and meteorological offices log rainfall at various locations. When data are collected at regular intervals, they form a time series. A historical time series is created from observations recorded at fixed intervals. In this context, time series are often treated as realizations of sequences of random variables, known as discrete-time stochastic processes or time series models. Our focus will be on applying these models using R to fit data and perform analysis.

Time series data often exhibit trends and seasonal variations that can be modeled mathematically. Additionally, observations close in time are typically correlated. Time series analysis aims to explain this correlation and other data features using statistical models. Once a model is fitted, it can be used to forecast future values, conduct statistical tests, and summarize the main characteristics of the data, aiding decision-making.Sampling intervals impact data quality. Aggregated data, like daily tourist arrivals, or sampled data, such as daily stock prices, need appropriate intervals to accurately reflect the original signal. In high-frequency trading or signal processing, continuous signals are sampled at very high rates to create time series for detailed analysis.

1.2.1 Uses of Time series

The analysis of time series is of great significance not only to economists and business people but also to scientists, astronomers, geologists, sociologists, biologists, and researchers. This is due to the following reasons:

- It helps in understanding past behavior.
- It assists in planning future operations.
- It aids in evaluating current accomplishments.
- It facilitates comparison.

1.2.2 Plots

Visualizing time series data is crucial for identifying patterns and trends. Common types of plots include:

- **Line Plot:** Displays data points connected by lines to show changes over time. Useful for identifying trends and seasonal patterns.
- **Scatter Plot:** Plots individual data points to observe the relationship between two variables or to identify patterns and outliers.
- **Bar Plot:** Represents data with bars, helpful for comparing discrete time periods or categories.
- **Histogram:** Shows the distribution of data over specified intervals, useful for understanding the frequency of values.
- **Box Plot:** Displays the distribution of data based on quartiles, highlighting median, and potential outliers.

1.2.3 Trends

Trends refer to the long-term movement or direction in the data over a period. Identifying trends helps in understanding the overall pattern:

- **Upward Trend:** Indicates a general increase in values over time.
- **Downward Trend:** Shows a general decrease in values.
- **Stationary Trend:** The data fluctuates around a constant mean without a long-term trend.

1.2.4 Seasonal Variation

Variations refer to the deviations from the trend and can be categorized into:

- **Seasonal Variations:** Regular patterns that repeat at consistent intervals, such as monthly or quarterly.
- **Cyclical Variations:** Fluctuations that occur over longer periods, influenced by economic or business cycles.
- **Irregular Variations:** Unpredictable changes due to unforeseen events or anomalies that do not follow a pattern.

Understanding these components allows for effective analysis and forecasting of time series data.

1.3 Decomposition of Series

1.3.1 Notation

The analysis so far has focused on plotting data to identify features such as trends and seasonal variations. While this is a crucial first step, the next stage involves fitting time series models. We represent a time series of length n as $\{x_t : t = 1, \ldots, n\} = \{x_1, x_2, \ldots, x_n\}$, where n values are sampled at discrete times $t = 1, 2, \ldots, n$. When the series length is not essential, we abbreviate it as $\{x_t\}$.

A time series model is a sequence of random variables, and the observed series is a realization of this model. We use the same notation for both, with context distinguishing between them. An overline denotes sample means.

$$\bar{x} = \sum \frac{x_i}{n} \tag{1}$$

The 'hat' notation represents a prediction or forecast. For a series $\{x_t : t = 1, \ldots, n\}$, $\hat{x}_{t+k|t}$ denotes a forecast made at time t for the value at $t + k$. The number of steps into the future, k, is the lead time. Depending on the context, $\hat{x}_{t+k|t}$ may refer to either the random variable or its numerical value.

1.3.2 Models

Many time series are dominated by trend and/or seasonal effects. A simple additive decomposition model is given by:

$$x_t = m_t + s_t + z_t \tag{2}$$

where x_t is the observed series, m_t is the trend, s_t is the seasonal effect, and z_t is the error term, often a sequence of correlated random variables with mean zero. Two main approaches for extracting m_t and s_t will be outlined along with R functions for this.

For cases where the seasonal effect increases with the trend, a multiplicative model may be more suitable:

$$x_t = m_t \cdot s_t + z_t \tag{3}$$

Alternatively, an additive decomposition for $\log(x_t)$ can be used:

$$\log(x_t) = m_t + s_t + z_t \tag{4}$$

Care is needed when transforming back to x_t from $\log(x_t)$ to avoid bias. If z_t is normally distributed with mean 0 and variance σ^2, the predicted mean value is:

$$\hat{x}_t = e^{m_t + s_t + \frac{1}{2}\sigma^2} \tag{5}$$

For non-normal distributions, bias correction may lead to overcorrection, requiring an empirical adjustment. This is critical, for instance, in financial forecasts, where underestimating mean costs is a common issue.

1.3.3 Estimating trends and seasonal effects

A simple way to estimate the trend m_t is by calculating a moving average centered on x_t. A moving average smooths the time series by averaging a specified number of values around each x_t, except for the first and last few terms. For monthly data, the moving average spans 12 months. Since the average of $t = 1$ (January) to $t = 12$ (December) falls between June and July (i.e., $t = 6.5$), we average two consecutive moving averages to center the result at $t = 7$. The centered moving average for m_t is given by:

$$\hat{m}_t = \frac{1}{12}\left(\frac{1}{2}x_{t-6} + x_{t-5} + \cdots + x_{t+5} + \frac{1}{2}x_{t+6}\right) \tag{6}$$

where $t = 7, \ldots, n-6$. The coefficients sum to 1, ensuring equal weight for each value. This method generalizes to other seasonal frequencies (e.g., quarterly) by maintaining the condition that coefficients sum to unity.

The seasonal effect $\hat{s}_t$ can be estimated by subtracting the trend:

$$\hat{s}_t = x_t - \hat{m}_t \tag{7}$$

Averaging the monthly estimates across all years provides a single estimate of the effect for each month. To ensure the seasonal effects sum to zero, they are adjusted by subtracting the mean. For multiplicative models, the estimate becomes:

$$\hat{s}_t = \frac{x_t}{\hat{m}_t} \tag{8}$$

and multiplicative factors are adjusted to average to 1. Seasonally adjusted data, often used in economic indicators, removes seasonal effects. If the seasonal effect is additive, the adjusted series is $x_t - \bar{s}_t$, and if multiplicative, it is $x_t/\bar{s}_t$, where $\bar{s}_t$ is the mean seasonal adjustment for the given time.

1.3.4 Smoothing

The centred moving average is a smoothing procedure applied retrospectively to identify an underlying trend in a time series. It uses points before and after the target time, often leaving some missing values at the series' start and end unless adapted for edge points. Another smoothing method in R is 'stl', which uses locally weighted regression (loess). This local regression considers a small number of points around the target time, weighted to reduce the influence of outliers, making it a robust regression. While straightforward in principle, the details of 'stl' are complex.

Unlike smoothing, which does not provide a forecast model, fitting a linear trend has the advantage of enabling extrapolation. The term "filtering" is also used in this context, particularly in engineering, to describe obtaining the best estimate of a variable based on past and current noisy measurements. Filtering is vital in control algorithms, such as those used by the Huygens probe during its 2005 landing on Titan.

1.3.5 Decomposition in *R*

In R, the function 'decompose' estimates trends and seasonal effects using a moving average. Nesting it within 'plot' (e.g., 'plot(stl())') produces a figure showing the original series x_t, and decomposed series m_t, s_t, and z_t. For example, additive and multiplicative decomposition plots for electricity data are created by the following commands, with the seasonal effect superimposed on the trend using 'lty' for line types.

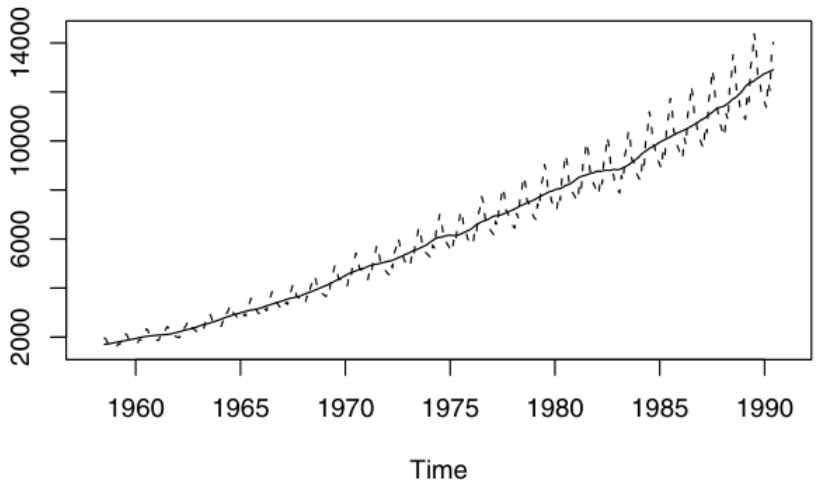


Figure 1: Electricity production data: trend with superimposed multiplicative seasonal effects.

```r
# Decomposition of the time series
plot(decompose(Elec.ts))

# Multiplicative decomposition
Elec.decom <- decompose(Elec.ts, type = "mult")
plot(Elec.decom)

# Extracting the trend and seasonal components
Trend <- Elec.decom$trend
Seasonal <- Elec.decom$seasonal

# Plotting the trend and the product of trend and seasonal effect
ts.plot(cbind(Trend, Trend * Seasonal), lty = 1:2)
```

Listing 1: Decomposition of Time Series in R

A multiplicative model is often more suitable than an additive one when the variance of the series and trend increase over time. However, if the random component z_t also shows increasing variance, a log-transformation (Eq. 1.4) may be more appropriate.

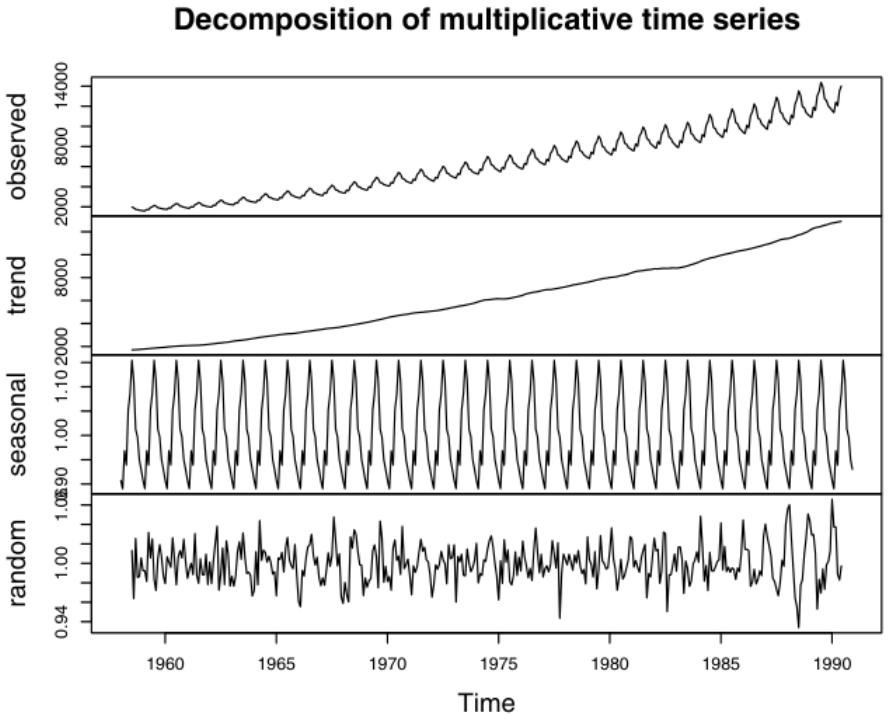


Figure 2: Decomposition of the electricity production data.

The random series from 'decompose' is not the true realization of z_t, but an estimate derived from the trend and seasonal components, treated as a residual error series, yet used as a realisation of the random process.

Module - 1

Chapter - 2

2 Characteristics of Time Series

A time series is a collection of data points indexed in time order, usually spaced at uniform intervals. It captures observations at successive points in time, and this ordering makes time series distinct from other types of data because time series analysis is inherently about trends, patterns, and dependencies across time. To effectively analyze time series data, it is important to understand its core characteristics. These characteristics form the basis for any meaningful analysis, forecasting, or modeling.

2.1 Introduction and Examples

Time series is a sequence of data points collected or recorded at regular time intervals. It is used in various domains such as economics, finance, meteorology, medicine, and engineering to analyze trends, patterns, and seasonal variations. Unlike cross-sectional data, time series data captures the dynamics and changes over time, allowing for forecasting and insight extraction from historical patterns.

Common examples of time series data include:

- Stock market prices recorded every minute.
- Daily temperature recordings in a city.
- Monthly sales data for a retail store.
- Quarterly GDP of a country.
- Yearly rainfall data in a specific region.

In R, we can visualize a simple time series data using the `AirPassengers` dataset:

```r
# Example in R:
data("AirPassengers")
plot(AirPassengers, main="AirPassengers Dataset",
     ylab="Number of Passengers", xlab="Year")
```

Listing 2: Example in R

2.2 Objectives and Nature of Time Series

The main objectives of time series analysis include:

- **Understanding** the underlying patterns in the data.
- **Identifying** components such as trend, seasonality, and cyclic behaviors.
- **Modeling** the time series to predict future values.
- **Detecting anomalies** or unexpected changes.
- **Smoothing** to eliminate noise and reveal important patterns.

The nature of time series can be categorized into:

- **Trend Component (T)**: A long-term increase or decrease in the data. For example, the overall upward movement in stock market prices over several years.

- **Seasonal Component (S)**: Regular fluctuations that repeat over a specific period, such as monthly sales peaking every December.
- **Cyclic Component (C)**: Recurrent but non-periodic fluctuations often linked to economic cycles.
- **Irregular Component (I)**: Random variations that do not follow a pattern.

These components can be combined using the following additive or multiplicative models:

$$Y(t) = T(t) + S(t) + C(t) + I(t) \tag{9}$$

or

$$Y(t) = T(t) \times S(t) \times C(t) \times I(t) \tag{10}$$

In R, we can decompose a time series to analyze these components:

```r
# Decomposition Example in R:
decomposed <- decompose(AirPassengers)
plot(decomposed)
```

Listing 3: Decomposition Example in R

2.3 Introduction to Time Series Databases and Applications

Time series databases (TSDB) are optimized for storing and querying time series data. Unlike traditional relational databases, TSDBs are built to efficiently handle timestamped data, which makes them ideal for applications that require storing large volumes of time-dependent data with high write-throughput and quick retrieval.

Some popular time series databases include:

- **InfluxDB**: InfluxDB is an open-source time series database designed specifically for storing and managing time series data such as metrics, events, and analytics. It is widely used in the Internet of Things (IoT), DevOps, real-time analytics, and monitoring applications. InfluxDB is part of the InfluxData stack, which includes Telegraf (a plugin-based collector of metrics), Chronograf (a visualization tool), and Kapacitor (an alerting and processing tool).

  InfluxDB uses a simple and flexible query language called InfluxQL, which is similar to SQL but optimized for time series operations like aggregations, time windowing, and transformations. It also supports the newer Flux language, which offers more powerful queries. InfluxDB is known for its high write throughput, enabling it to handle millions of writes per second, which makes it ideal for real-time data collection and analysis.

  InfluxDB stores data in a compressed time series format and offers a retention policy feature, where users can automatically delete old data to manage storage effectively. It also supports downsampling, which helps reduce storage costs by keeping high-resolution data for shorter periods while retaining summarized, lower-resolution data for longer periods.

  InfluxDB uses the InfluxQL or Flux query languages. Below is an example using InfluxQL.

```
# Storing time series data in InfluxDB
INSERT temperature,location=room1 value=72.5 1627550220
INSERT temperature,location=room2 value=75.3 1627550280
```

Listing 4: Storing Time Series Data in InfluxDB

```
# Querying time series data in InfluxDB
SELECT mean("value") FROM "temperature"
WHERE time >= '2021-07-29T00:00:00Z' AND time < '2021-07-30T00:00:00Z'
GROUP BY time(1h), "location" fill(none)
```

Listing 5: Querying Time Series Data in InfluxDB

Advantages of InfluxDB:

- **High Performance**: InfluxDB is optimized for fast writes, making it ideal for use cases where data is collected frequently, such as IoT applications or system monitoring. Its write throughput is among the best in class for time series databases.
- **Retention Policies**: Users can define retention policies to automatically expire older data, thus managing storage costs efficiently. This is particularly useful in environments where data grows exponentially over time.
- **Schema-Free**: InfluxDB is schemaless, meaning that data can be written with any fields and tags, making it flexible to adapt to new use cases and metrics without predefined structures.
- **Integrations**: InfluxDB integrates easily with other tools such as Grafana for visualization, Telegraf for data collection, and Kapacitor for alerting and data processing.

Cons of InfluxDB:

- **Query Complexity**: While InfluxQL is simple for basic queries, more complex queries involving joins or transformations might be challenging. Flux, the newer query language, addresses these issues but introduces a learning curve.
- **Scaling Issues**: Scaling InfluxDB horizontally (i.e., across multiple nodes) can be challenging. The enterprise version of InfluxDB offers clustering, but the open-source version does not, making scaling limited for high-availability deployments.
- **Storage Costs**: Although InfluxDB offers compression, the storage requirements for high-frequency data can still be substantial, especially in long-term retention scenarios.

- **TimescaleDB**: TimescaleDB is an open-source time series database built as an extension to PostgreSQL. By leveraging the mature and robust PostgreSQL ecosystem, TimescaleDB inherits features like ACID compliance, powerful indexing, relational joins, and SQL support, making it a reliable option for time series applications that also require relational data.

  TimescaleDB splits large datasets into "chunks" based on time intervals, which allows for efficient time-series queries. This method also makes TimescaleDB horizontally scalable while keeping storage costs down. Additionally, TimescaleDB supports native compression, significantly reducing the storage footprint for large datasets.

  One of the strengths of TimescaleDB is its seamless integration with the broader PostgreSQL ecosystem, which includes extensions, tools, and libraries. This makes it an attractive choice for users who already have a PostgreSQL setup and are looking to incorporate time series capabilities into their existing infrastructure without migrating to a new system.

  TimescaleDB uses standard SQL, with time series data stored in hypertables.

```sql
-- Creating a hypertable in TimescaleDB
CREATE TABLE temperature (
    time        TIMESTAMPTZ       NOT NULL,
    location    TEXT              NOT NULL,
    temperature DOUBLE PRECISION  NOT NULL
);

-- Convert the table to a hypertable
SELECT create_hypertable('temperature', 'time');

-- Inserting data into TimescaleDB
INSERT INTO temperature (time, location, temperature)
VALUES (NOW(), 'room1', 72.5), (NOW(), 'room2', 75.3);
```

Listing 6: Storing Time Series Data in TimescaleDB

```sql
-- Querying the average temperature by hour
SELECT time_bucket('1 hour', time) AS bucket, location, avg(temperature)
FROM temperature
WHERE time > NOW() - interval '24 hours'
GROUP BY bucket, location
```

```
ORDER BY bucket DESC;
```

Listing 7: Querying Time Series Data in TimescaleDB

Advantages of TimescaleDB:

- **PostgreSQL Ecosystem**: Since TimescaleDB is an extension of PostgreSQL, it benefits from the stability, reliability, and community support of PostgreSQL. This includes support for advanced indexing, relational joins, transactions, and other features common to relational databases.
- **SQL Support**: TimescaleDB supports standard SQL queries, making it easy for developers familiar with SQL to work with time series data. This reduces the learning curve compared to other TSDBs that use custom query languages.
- **Efficient Time Series Storage**: TimescaleDB automatically partitions data into chunks based on time intervals, which improves query performance. It also supports data compression, making it highly efficient for storing large datasets.
- **Scalability**: TimescaleDB provides built-in tools for scaling horizontally, allowing it to handle large time series datasets across distributed environments.

Cons of TimescaleDB:

- **Limited for Extreme Real-Time Use Cases**: While TimescaleDB performs well for most time series applications, it may not be as optimized for extreme high-frequency, real-time applications as InfluxDB or Prometheus.
- **Complexity with Large Joins**: Although relational joins are a strength of TimescaleDB, performing large-scale joins on massive datasets can lead to performance issues, particularly for real-time queries.
- **Enterprise Features**: Some advanced features, like continuous aggregation and advanced compression, are part of TimescaleDB's enterprise offering, which can be a limitation for users relying only on the open-source version.

- **Prometheus**: Prometheus is a highly popular, open-source monitoring and alerting toolkit designed specifically for cloud-native environments. It was developed as part of the Cloud Native Computing Foundation and is often used in conjunction with Kubernetes for monitoring application performance, infrastructure metrics, and other system behaviors.

  Prometheus works by scraping metrics from instrumented services at regular intervals, storing them as time series data. It supports multi-dimensional data collection using labels, which are key-value pairs attached to the metrics. Prometheus uses its own query language called PromQL (Prometheus Query Language), which is specifically designed for aggregating and filtering time series data. Its alerting mechanism is flexible and integrates easily with various notification systems like PagerDuty, Slack, and email.

  One of the primary use cases for Prometheus is in monitoring cloud infrastructure, where it excels at tracking the performance of servers, containers, and microservices. The system is designed to be lightweight and works well in environments where quick real-time insights and monitoring are critical.

  Prometheus uses PromQL for querying, and data is scraped from instrumented services.

```
# An example of Prometheus scrape configuration
scrape_configs:
  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']
```

Listing 8: Scraping Time Series Data in Prometheus

```
# Querying time series data in Prometheus using PromQL
avg_over_time(node_cpu_seconds_total[5m])
```

Listing 9: Querying Time Series Data in Prometheus

Advantages of Prometheus:

- **Cloud-Native Friendly**: Prometheus is designed to work seamlessly in dynamic cloud-native environments, particularly with Kubernetes. It is well-suited for containerized environments where services come and go frequently.
- **Highly Scalable**: Prometheus is built for large-scale, distributed environments. Its pull-based metrics collection makes it efficient for monitoring hundreds or thousands of services.
- **Alerting System**: Prometheus has a powerful alerting system that allows users to define alerting rules based on metric thresholds. It integrates easily with notification tools, enabling quick responses to system failures or abnormal metrics.
- **Multi-Dimensional Data Collection**: Prometheus allows users to attach labels to their metrics, making it easy to filter and aggregate data along different dimensions, such as by service, data center, or cluster.

Cons of Prometheus:

- **Limited Long-Term Storage**: Prometheus is not designed for long-term data retention. While it excels at real-time monitoring, users often need to integrate it with external databases like Thanos or Cortex to store time series data for long-term historical analysis.
- **No Built-In Clustering**: Prometheus does not support clustering in its native form, which can be a limitation for users requiring high availability and fault tolerance without external dependencies.
- **Query Language Complexity**: PromQL, the query language used by Prometheus, can be complex for new users, particularly for those used to SQL or other more common query languages. Learning to write efficient queries in PromQL can take time.

Applications of time series databases include:

- **IoT Data Storage**: Time series databases are commonly used in IoT devices to store sensor data, such as temperature readings, GPS data, or humidity levels.
- **Financial Market Analysis**: TSDBs handle high-frequency trading data, storing stock prices, trading volumes, and other financial indicators over time.
- **DevOps Monitoring**: Tracking system performance metrics like CPU usage, memory consumption, and network bandwidth usage in real-time.

An R example to work with a simple time series dataset:

```r
# Simulate time series data and store it
time_series_data <- ts(rnorm(100), frequency=12, start=c(2020, 1))
plot(time_series_data, main="Simulated Time Series Data",
     ylab="Values", xlab="Time")
```

Listing 10: Simulate Time Series Data in R

2.4 Measures of Dependence

2.4.1 Introduction to Measures of Dependence

In time series analysis, *measures of dependence* refer to the statistical relationships between observations in a time series dataset, especially over time lags. Understanding these dependencies is crucial because they indicate whether and how past values influence future values. A time series is dependent when the values at different points in time are not independent but rather exhibit a relationship that we can measure and analyze.

One of the primary measures of dependence in time series is the **autocorrelation function** (ACF). This function helps determine how observations at different time points are correlated with one another. The ACF for a time series $\{X_t\}$ at lag k is given by:

$$\rho(k) = \frac{\text{Cov}(X_t, X_{t+k})}{\sqrt{\text{Var}(X_t) \cdot \text{Var}(X_{t+k})}}$$

where Cov denotes covariance, and Var represents the variance of the time series.

The partial autocorrelation function (PACF) is another measure of dependence, which describes the relationship between an observation and its lagged values, excluding the influence of intermediate lags. This is particularly useful in autoregressive (AR) models, where PACF helps determine the order of the model.

2.4.2 Example Problem: Estimating Autocorrelation

Consider a simple time series dataset where we observe daily stock prices for 10 days:

$$X = \{120, 122, 119, 118, 121, 123, 124, 125, 126, 128\}.$$

We want to calculate the autocorrelation at lag 1 and lag 2.

Step 1: Compute the mean of the series. First, compute the mean $\bar{X}$ of the series:

$$\bar{X} = \frac{120 + 122 + 119 + 118 + 121 + 123 + 124 + 125 + 126 + 128}{10} = \frac{1226}{10} = 122.6.$$

Step 2: Define the autocorrelation formula. The formula for autocorrelation at lag k is:

$$\rho(k) = \frac{\sum_{t=1}^{n-k}(X_t - \bar{X})(X_{t+k} - \bar{X})}{\sum_{t=1}^{n}(X_t - \bar{X})^2},$$

where:

- X_t is the value at time t,
- $\bar{X}$ is the mean of the time series,
- k is the lag,
- n is the total number of observations.

Step 3: Calculate the denominator for all lags. The denominator for both lag 1 and lag 2 is the same:

$$\sum_{t=1}^{n}(X_t - \bar{X})^2 = (120 - 122.6)^2 + (122 - 122.6)^2 + (119 - 122.6)^2 + \ldots + (128 - 122.6)^2.$$

Substitute the values:

$$= (-2.6)^2 + (-0.6)^2 + (-3.6)^2 + (-4.6)^2 + (-1.6)^2 + (0.4)^2 + (1.4)^2 + (2.4)^2 + (3.4)^2 + (5.4)^2.$$

$$= 6.76 + 0.36 + 12.96 + 21.16 + 2.56 + 0.16 + 1.96 + 5.76 + 11.56 + 29.16 = 92.4.$$

Thus, the denominator is 92.4.

Step 4: Calculate the numerator for lag 1. Now, compute the numerator for lag 1, which is:

$$\sum_{t=1}^{n-1}(X_t - \bar{X})(X_{t+1} - \bar{X}) = (120 - 122.6)(122 - 122.6) + (122 - 122.6)(119 - 122.6) + \ldots + (126 - 122.6)(128 - 122.6).$$

Substitute the values:

$$= (-2.6)(-0.6) + (-0.6)(-3.6) + (-3.6)(-4.6) + (-4.6)(-1.6) + (-1.6)(0.4) + (0.4)(1.4) + (1.4)(2.4) + (2.4)(3.4) + (3.4)(5.4).$$

$$= 1.56 + 2.16 + 16.56 + 7.36 + (-0.64) + 0.56 + 3.36 + 8.16 + 18.36 = 57.68.$$

Step 5: Calculate autocorrelation at lag 1. Now that we have the numerator and denominator, calculate the autocorrelation:

$$\rho(1) = \frac{57.68}{92.4} \approx 0.624.$$

Thus, the autocorrelation at lag 1 is approximately 0.624, indicating a moderate positive correlation between consecutive values.

Step 6: Calculate the numerator for lag 2. For lag 2, compute the numerator:

$$\sum_{t=1}^{n-2}(X_t - \bar{X})(X_{t+2} - \bar{X}) = (120-122.6)(119-122.6) + (122-122.6)(118-122.6) + \ldots + (125-122.6)(128-122.6).$$

Substitute the values:

$$= (-2.6)(-3.6) + (-0.6)(-4.6) + (-3.6)(-1.6) + (-4.6)(0.4) + (-1.6)(1.4) + (0.4)(2.4) + (1.4)(3.4) + (2.4)(5.4).$$

$$= 9.36 + 2.76 + 5.76 + (-1.84) + (-2.24) + 0.96 + 4.76 + 12.96 = 32.48.$$

Step 7: Calculate autocorrelation at lag 2. Finally, calculate the autocorrelation at lag 2:

$$\rho(2) = \frac{32.48}{92.4} \approx 0.352.$$

Thus, the autocorrelation at lag 2 is approximately 0.352, indicating a weaker positive correlation between values that are two time steps apart.

Conclusion From these calculations, we see that the autocorrelation at lag 1 is higher (0.624) compared to lag 2 (0.352). This suggests that consecutive stock prices are more closely related than prices separated by two days, which is a typical observation in time series where immediate past values have a stronger influence on the present.

2.5 Stationary Time Series

2.5.1 Definition and Importance of Stationarity

A *stationary time series* is one whose statistical properties such as mean, variance, and autocorrelation are constant over time. Stationarity is important because many time series models, such as ARMA, ARIMA, or GARCH, assume that the data is stationary. If the data is not stationary, the model's performance can suffer, leading to poor forecasts.

Formally, a time series $\{X_t\}$ is stationary if for all time points t, the following conditions hold:

- $\mathbb{E}(X_t) = \mu$ (constant mean)
- $\text{Var}(X_t) = \sigma^2$ (constant variance)
- $\text{Cov}(X_t, X_{t+k}) = \gamma(k)$ (constant autocovariance that depends only on lag k)

2.5.2 Features of Stationary Time Series

Key characteristics of a stationary series include:

- The series fluctuates around a constant mean.
- There is no long-term trend.
- The autocorrelation function decreases quickly as the lag increases.

2.5.3 R Example: Plotting a Stationary Series

We can generate and plot a stationary time series in R using the following code:

```r
set.seed(123)
stationary_series <- ts(rnorm(100), frequency=12)
plot(stationary_series, main="Stationary Time Series", ylab="Values", xlab="Time")
```

Listing 11: Generating a Stationary Time Series

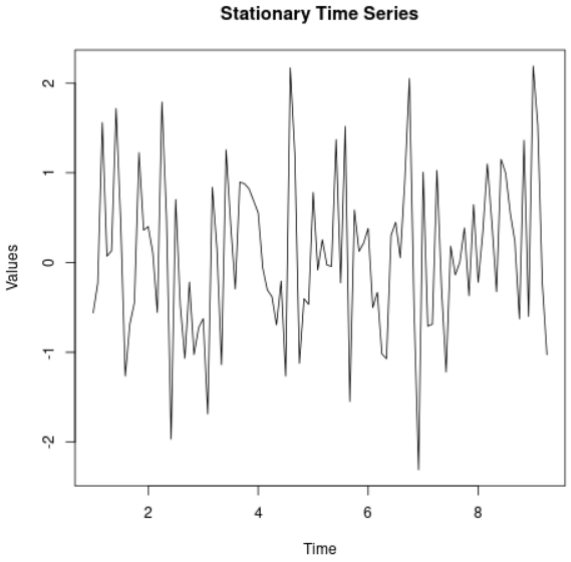


Figure 3: Stationary Time Series

The plot of the stationary series will show random fluctuations around a constant mean with no visible trend.

2.6 Estimation of Correlation

2.6.1 Definition of Correlation

Correlation is a measure of the strength and direction of the linear relationship between two variables. In the context of time series, correlation helps identify how strongly values at one time point relate to values at a later time point (lagged values). Estimating correlation is crucial for understanding the underlying patterns in the data, such as seasonality or trends.

The Pearson correlation coefficient is given by:

$$r = \frac{\sum_{t=1}^{n}(X_t - \bar{X})(Y_t - \bar{Y})}{\sqrt{\sum_{t=1}^{n}(X_t - \bar{X})^2 \sum_{t=1}^{n}(Y_t - \bar{Y})^2}}$$

In time series analysis, we often compute *autocorrelation* or the correlation between values at different time lags. Estimating the autocorrelation function (ACF) helps us determine whether previous values have predictive power for future values.

2.6.2 Proof of Correlation for Time Series

For a time series $\{X_t\}$, the autocorrelation at lag k is:

$$\rho(k) = \frac{\sum_{t=k+1}^{n}(X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^{n}(X_t - \bar{X})^2}$$

This equation calculates the correlation between observations separated by k time steps. As k increases, $\rho(k)$ typically decreases, reflecting the diminishing influence of earlier observations on future values.

2.7 Vector-Valued and Multi-Dimensional Series

2.7.1 Definition and Importance

A *vector-valued time series* consists of multiple time series observed together. These can be considered multidimensional or multivariate, where each dimension represents a different but related time series. Analyzing such series is

important in fields like economics (e.g., analyzing stock prices for multiple companies) and environmental science (e.g., temperature, humidity, and wind speed together).

Multidimensional time series analysis focuses on understanding the relationships between these multiple series and how they jointly evolve over time. The vector autoregressive (VAR) model is a common model used for such series.

2.7.2 Example: Vector-Valued Series in R

We can create and analyze a vector-valued time series in R using the following code:

```r
# Simulate two related time series
set.seed(123)
ts1 <- ts(rnorm(100), frequency=12)
ts2 <- ts(rnorm(100, mean=2), frequency=12)

# Combine them into a multivariate time series
multi_series <- ts(cbind(ts1, ts2), frequency=12)

# Plot the multivariate series
plot(multi_series, main="Vector-Valued Time Series", col=c("blue", "red"), lty=1:2)
legend("topright", legend=c("Series 1", "Series 2"), col=c("blue", "red"), lty=1:2)
```

Listing 12: Creating and Plotting a Multidimensional Time Series

Equation for Multivariate Model: In a multivariate time series model, each variable depends on its own past values and the past values of other variables. The vector autoregressive (VAR) model for two time series X_t and Y_t is given by:

$$X_t = \alpha_1 X_{t-1} + \beta_1 Y_{t-1} + \epsilon_{1t}$$
$$Y_t = \alpha_2 X_{t-1} + \beta_2 Y_{t-1} + \epsilon_{2t}$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are coefficients and $\epsilon_{1t}, \epsilon_{2t}$ are error terms.

This type of modeling is crucial in understanding how multiple series interact over time.

Module - 1

Chapter - 3 & 4

3 Components of Time Series

article amsmath

The characteristics of a time series are defined by the various types of movements or fluctuations that occur over time. These movements, known as the components of a time series, help explain the underlying patterns in the data. There are four main components:

1. Secular Trend (T)

The **Secular Trend**, also known as the **long-term trend** or simply **trend**, refers to the general movement of data, either upward or downward, over an extended period. It captures the long-term tendency of a dataset to grow or decline, ignoring short-term fluctuations.

For example, the population of India shows a clear upward trend over the years, while the death rate after independence has steadily declined due to improvements in literacy and healthcare. It's important to note that what constitutes a "long period" depends on the context of the data. For instance, an increase in cloth store sales over one year (e.g., from 1996 to 1997) is too short a period to be considered a secular trend.

However, in certain cases, a shorter time frame can reflect a trend if the nature of the data allows. For example, in a bacterial culture exposed to germicide, counting the number of organisms still alive every 10 seconds over 5 minutes could reveal a general decline in numbers, which would represent a secular trend over that period.

Mathematically, secular trends are categorized into two types:

1. **Linear Trend**: A consistent, straight-line increase or decrease over time.
2. **Curvi-Linear Trend (Non-Linear Trend)**: A trend where the rate of change is not constant, resulting in a curved pattern.

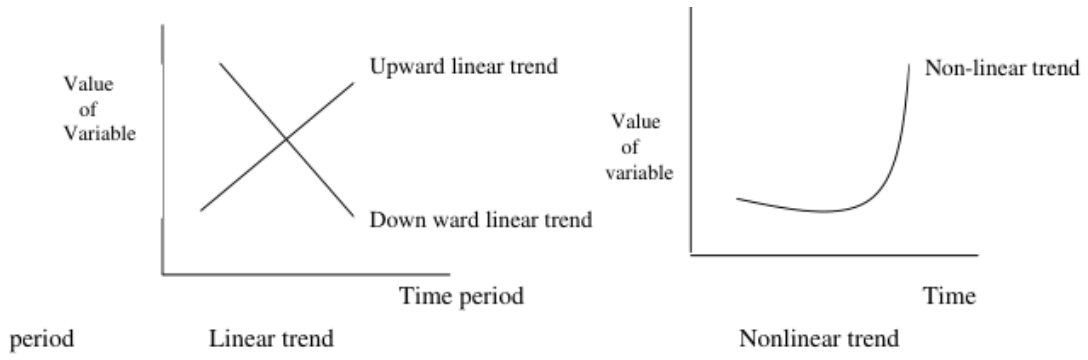


Figure 4: Linear Trend and Non Linear Trend

Example: A consistent rise in global temperatures over decades due to climate change.

2. Seasonal Variations (S)

Seasonal variations occur in a time series due to rhythmic forces that repeat in a regular and periodic manner within a period of less than one year. These variations follow the same pattern year after year. The period may be monthly, weekly, or even hourly, but if data is given in yearly terms, seasonal fluctuations do not exist.

Seasonal fluctuations in a time series arise from two main factors:

1. **Natural forces**
2. **Manmade conventions**

The most significant cause of seasonal variations is climate. Changes in weather conditions—such as rainfall, humidity, and temperature—impact industries and products differently. For example, there is a higher demand for woolen clothes and hot drinks in winter, while in summer, cotton clothes and cold drinks see increased sales. During the rainy season, the demand for umbrellas and raincoats rises.

In addition to nature, customs, traditions, and habits also influence seasonal variation. For instance, during festivals like Diwali, Dussehra, and Christmas, there is an increased demand for sweets and clothes. Similarly, the start of a school or college year sees a surge in demand for books and stationery.

Example: Higher sales of air conditioners during summer months due to the hot weather.

3. Cyclical Variations (C)

Cyclical movements occur over longer time periods than seasonal variations and typically reflect economic cycles such as booms and recessions. These cycles generally last for several years and, unlike seasonal variations, they do not follow a fixed or regular pattern.

Cyclical variations refer to short-term fluctuations lasting more than one year. The rhythmic movements in a time series that repeat in the same manner over a period longer than one year are called cyclical variations, and the duration is referred to as a **cycle**. Time series related to business and economics often exhibit cyclical behavior.

A classic example of cyclical variation is the **Business Cycle**, which includes four well-defined phases:

1. **Boom**: This phase is characterized by rapid economic growth, high levels of production, employment, and rising prices. During the boom period, consumer demand is strong, and businesses expand rapidly. However, inflationary pressures may also build up, leading to potential overheating of the economy.

   Example: The global economy in the late 1990s experienced a boom due to the dot-com bubble, where technology companies saw rapid growth and expansion.

2. **Decline**: After the boom, the economy begins to slow down. Production and demand decrease, unemployment starts to rise, and inflation stabilizes. This phase marks the transition from a peak towards a downturn, signaling the end of rapid economic expansion.

   Example: The early 2000s saw a decline after the burst of the dot-com bubble, where stock prices fell, and many tech companies collapsed, leading to a slowdown in economic growth.

3. **Depression**: This is the lowest phase of the cycle, marked by a significant decline in economic activity. There is high unemployment, reduced consumer spending, lower investment, and overall economic stagnation. It represents the most severe form of economic contraction.

   Example: The Great Depression of the 1930s is a classic example, where global economies shrank, unemployment reached record levels, and industrial output dropped sharply.

4. **Improvement (Recovery)**: After the depression, the economy begins to recover. Businesses start investing again, employment rises, and consumer confidence gradually returns. Production and demand start increasing, marking the beginning of the next upward cycle.

   Example: After the Great Recession of 2008, the economy began recovering in 2010, with improved job growth, increased consumer spending, and steady economic expansion.

These phases repeat over time, reflecting the fluctuating nature of economic activity.

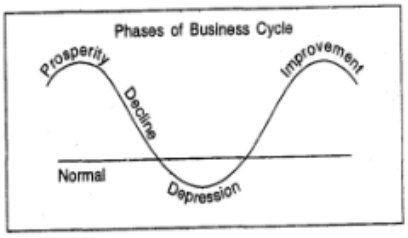


Figure 5: Phases of Business cycle

Example: Economic cycles with alternating periods of economic expansion and contraction.

4. Irregular Variations (I)

Irregular variations, also known as **Erratic**, **Accidental**, or **Random Variations**, are unpredictable and non-recurring fluctuations in a time series caused by unexpected events. Unlike trend, seasonal, and cyclical variations—which are considered regular variations—irregular variations are random and typically short-term, making them difficult to model or forecast.

These fluctuations are the result of unforeseen circumstances that are beyond human control, such as natural disasters, wars, pandemics, or other catastrophic events. Irregular variations significantly disrupt a time series but are not as structurally important as other variations.

Example: The COVID-19 pandemic in 2020 led to severe and unexpected disruptions across global economies, causing irregular variations in many time series related to employment, GDP, and stock market performance. This variation could not be predicted and doesn't follow any consistent or repeating pattern.

Together, these components provide a framework to analyze time series data, enabling better forecasting and understanding of the underlying patterns.

3.1 Additive and Multiplicative models

In time series analysis, a **mathematical model** represents the underlying structure of the data. It is assumed that the time series consists of various components such as trends, seasonal variations, cyclical variations, and irregular variations. These components together explain the observed value of the time series at any point in time.

The objective of a mathematical model is to decompose a time series into its constituent components in order to better understand, analyze, and forecast future values. Two widely used models in classical time series analysis are the **Additive Model** and the **Multiplicative Model**.

Why Are Mathematical Models Needed?

Mathematical models are essential in time series analysis for the following reasons:

1. **Understanding patterns**: By decomposing the time series into its components, we can identify trends, seasonal behaviors, and cyclical movements that help in understanding the nature of the data.
2. **Forecasting**: Mathematical models help us forecast future values based on past observations and the relationships among the components.
3. **Analyzing irregularities**: With the model, irregular or random variations can be separated from systematic variations, allowing analysts to focus on predictable aspects of the data.

Additive Model

The **Additive Model** assumes that the different components of a time series combine in an additive manner. That is, the observed value Y_t at time t is the sum of the contributions of the individual components.

Mathematically, the additive model is represented as:

$$Y_t = T_t + S_t + C_t + I_t$$

where:

- Y_t is the observed value at time t,
- T_t is the **trend component** at time t,
- S_t is the **seasonal component** at time t,
- C_t is the **cyclical component** at time t, and
- I_t is the **irregular component** at time t.

Derivation of Additive Model

The additive model is useful when the variation in the seasonal and cyclical components remains relatively constant over time. For example, if sales of ice cream increase by a fixed amount every summer, we can model that seasonal variation additively.

Example

Consider a time series of monthly sales data for a store over a year. Suppose the trend increases by 5 units per month, the seasonal effect adds 10 units during the summer months (June, July, and August), and cyclical factors add or subtract up to 3 units. Then, using the additive model, we can express the sales data Y_t for a summer month as:

$$Y_t = 5t + 10 + C_t + I_t$$

where C_t is the cyclical effect and I_t represents any irregular variations.

Multiplicative Model

The **Multiplicative Model** assumes that the components of the time series interact in a multiplicative manner. That is, the observed value Y_t at time t is the product of the contributions of the individual components.

Mathematically, the multiplicative model is represented as:

$$Y_t = T_t \times S_t \times C_t \times I_t$$

where the variables Y_t, T_t, S_t, C_t, and I_t represent the same components as in the additive model.

Derivation of Multiplicative Model

The multiplicative model is useful when the seasonal and cyclical variations are proportional to the level of the trend. For instance, if sales of ice cream double in the summer but are still dependent on an overall increasing trend, a multiplicative model would be more appropriate.

Example

Consider a company's quarterly revenue over a few years. If the trend increases by 10% each quarter, and sales are doubled during the holiday season, the multiplicative model expresses the revenue as:

$$Y_t = T_t \times 2 \times C_t \times I_t$$

where T_t represents the 10% growth in each quarter, the factor 2 accounts for the seasonal holiday surge, C_t captures any cyclical effects, and I_t represents irregular variations.

Choosing Between Additive and Multiplicative Models

The choice between the additive and multiplicative models depends on the nature of the data:

- **Additive Model**: Appropriate when the variations are constant over time and do not depend on the trend.
- **Multiplicative Model**: Suitable when the variations grow or shrink in proportion to the trend.

Conclusion

Both the additive and multiplicative models provide valuable ways to decompose a time series into its underlying components. By choosing the right model, analysts can gain better insights into trends, seasonal variations, and cyclical movements, and make more accurate forecasts.

3.2 Resolving components of a Time Series

In time series analysis, resolving the different components is a fundamental task to understand the underlying patterns. Time series data is usually composed of several components, and the key components include:

- **Trend** (T_t): The long-term movement in the data over time.
- **Seasonality** (S_t): Regular patterns that repeat over fixed intervals of time.
- **Cyclicality** (C_t): Long-term fluctuations caused by economic cycles.
- **Irregularity** (I_t): Random or unpredictable movements, typically caused by unforeseen factors.

The relationship between these components can be expressed using two main models:

- **Additive Model:**
$$Y_t = T_t + S_t + C_t + I_t$$
- **Multiplicative Model:**
$$Y_t = T_t \times S_t \times C_t \times I_t$$

In R, you can resolve components of a time series using built-in functions like 'decompose()' for additive models or 'stl()' for both additive and multiplicative models. Consider the following example where we decompose the AirPassengers dataset.

```r
```r
# Load AirPassengers dataset
data(AirPassengers)

# Decompose the time series
decomposed_data <- decompose(AirPassengers, type = "multiplicative")

# Plot decomposed components
plot(decomposed_data)
```

Listing 13: Creating and Plotting a Multidimensional Time Series
```

3.3 Measuring Trend

The trend component of a time series reflects the long-term movement in the data. Understanding the trend is crucial for forecasting future values and identifying underlying patterns. There are several methods commonly used to measure trends:

3.3.1 Graphic

The graphic method, also known as the eye inspection method, is the simplest and most intuitive approach to identifying trends in time series data. This method involves the following steps:

1. **Plot the Data:** First, plot the given time series data on a graph, with time on the x-axis and the variable of interest on the y-axis.
2. **Draw a Trend Line:** A smooth, free-hand curve is then drawn through the plotted points, representing the general tendency of the series. This curve visually highlights the trend over time.

The graphic method effectively removes short-term variations to reveal the underlying trend in the data. The trend line can also be extended to predict or estimate future values, making it a useful tool for forecasting.

Importance of the Graphic Method

- Provides a visual and intuitive understanding of the trend.
- Easy to implement, requiring no complex calculations.
- Serves as a preliminary tool before applying more sophisticated methods.

Limitations

However, it is important to note that this method is subjective, and the accuracy of the predictions may vary depending on how the trend line is drawn. As such, while the graphic method is useful for initial analysis, it should be supplemented with more rigorous statistical techniques for reliable forecasting.

Example

Consider monthly sales data for a retail store over a year:

Month	Sales
Jan	100
Feb	120
Mar	140
Apr	160
May	150
Jun	130
Jul	180
Aug	190
Sep	170
Oct	160
Nov	140
Dec	200

Table 2: Monthly Sales Data

In R, the plotting can be done using the following code:

```r
# Define the sales data
sales <- c(100, 120, 140, 160, 150, 130, 180, 190, 170, 160, 140, 200)
months <- 1:12

# Plot the sales data
plot(months, sales, type = "o", col = "blue", xlab = "Month", ylab = "Sales")

# Add a manually drawn trend line (approximate)
lines(c(1, 12), c(100, 200), col = "red", lwd = 2)
```

Listing 14: Creating and Plotting a Multidimensional Time Series

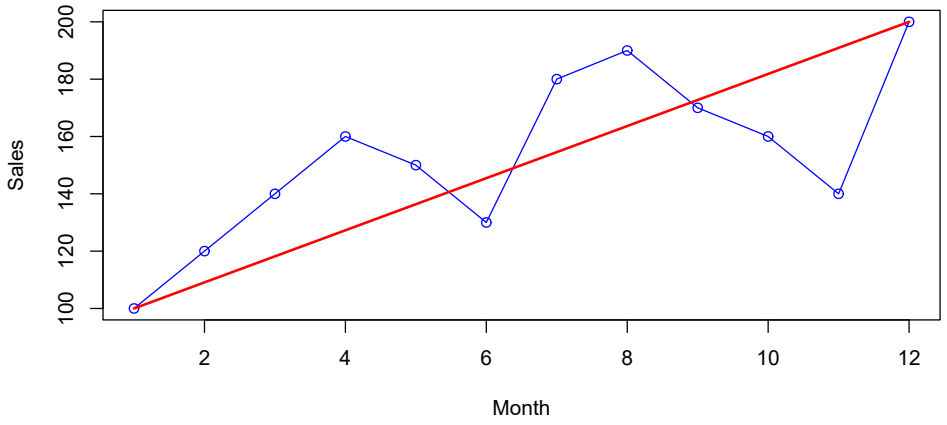


Figure 6: Trend from the Data

The red line represents the overall trend in sales. Although the data fluctuates, the general upward direction is clearly visible.

Advantages:

- **Simplicity:** The graphic method is one of the simplest approaches to studying trend values and is easy to implement.
- **Expertise Benefits:** An experienced statistician can often draw a trend line that better represents the data than one fitted using mathematical formulas.
- **Applicability:** Despite not being recommended for beginners, this method has significant merits in the hands of skilled statisticians and is widely used in practical applications.

Disadvantages:

- **Subjectivity:** The method is highly subjective; the resulting trend line can vary significantly based on who draws it.
- **Skill Requirements:** It requires the work to be conducted by skilled and experienced individuals to ensure accuracy.
- **Reliability Concerns:** The subjective nature of this method means that predictions derived from it may not be reliable.
- **Careful Execution:** Drawing the trend line must be done carefully to avoid misrepresentation of the data.